AGGREGATING LOGNORMAL RISKS IN GROUPS: JOINT TAILS AND COPULAS

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ABSTRACT. In this contribution we investigate simultaneous large values of sums and maxima of dependent lognormal risks partitioned into finitely many groups. We obtain sharp joint-tail asymptotics and show that the two tails are asymptotically proportional, extending the scalar single-big-jump principle. It turns out that the copulas of the group sums and maxima are asymptotically independent and have the same logarithmic upper-tail exponents. The joint extreme-value copula of the sums and maxima exhibits complete dependence within each group and independence between groups. Further, for two groups we derive a sharp copula-tail equivalent when Gaussian margins are identical within each group and establish hidden regular variation for common Gaussian variance and arbitrary means. We also obtain conditional aggregation and common-excess limits and extend tail proportionality to signed sums under nondegeneracy assumptions. A log-elliptical extension distinguishes finite and diverging conditional residual scales.

1. INTRODUCTION

Lognormal models are widely used to model positive risks subject to multiplicative variation. In insurance and finance, sums of dependent lognormal random variables arise in the modelling of aggregate losses and portfolio values, while weighted sums enter the pricing of arithmetic basket options [3, 10, 14]. The aggregation problem also has a long history in communication systems [12]. In wireless communications, sums of lognormal variables describe aggregate interference under lognormal shadowing [23, 29]. Lognormal models also occur in environmental science for pollutant concentrations [25] and in cellular biology for protein abundances across populations of bacterial cells [13]. These applications motivate the study of both individual lognormal quantities and their aggregates.

A basic result for a single sum (aggregate) is the single-big-jump principle. Specifically, for a Gaussian random vector $(X_1, \dots, X_n)$ with positive definite covariance matrix, [3] shows that

$$\mathbb{P}\left\{\sum_{j=1}^{n} e^{X_j} > u\right\} \sim \mathbb{P}\left\{\max_{1\le j\le n} e^{X_j} > u\right\}, \qquad u \to \infty.$$

In our notation $f(u) \sim g(u)$ for two positive functions f, g means $\lim_{u\to\infty} f(u)/g(u) = 1$.

The above approximation shows that a large value of the sum is asymptotically accounted for by its largest summand. Related aggregation results have been established for log-elliptical risks [17], suitable classes of log-linear risks [11] and weighted sums of powers of Dirichlet risks [18]; a general approach to the aggregation of rapidly varying risks is presented in [24].

The scalar principle also underlies efficient simulation of rare events for correlated lognormal sums [2]. For sharp tail asymptotics of linear combinations, including differences of correlated lognormal variables, see [14].

In many applications, dependent risks are reported or managed in groups. Examples include losses in different business lines, values of several portfolios and interference totals associated with different groups of transmitters. This leads to a vector of group sums and to questions about simultaneous large values of its components.

The classical scalar single-big-jump principle applies to each group separately. In order to assess simultaneous exceedances, one needs the joint tail of the group totals and its relation to the joint tail of the corresponding group maxima.

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Next we introduce our mathematical setting: Let $A_1,\ldots,A_d$ be a partition of $\{1,\ldots,N\}$ into nonempty sets, where N and d are fixed positive integers. Suppose that $\boldsymbol{X}$ is an $\mathbb{R}^N$-valued Gaussian random vector with mean vector $\boldsymbol{\mu}$ and positive definite covariance matrix Σ, i.e.,

$$\boldsymbol{X}\sim \mathcal{N}_N(\boldsymbol{\mu},\Sigma),\qquad \Sigma\succ 0 \tag{1.1}$$

and define the corresponding group sums and group maxima by

$$\boldsymbol{S}=\left(\sum_{j\in A_\ell} e^{X_j}\right)_{1\le \ell\le d},\qquad \boldsymbol{Q}=\left(\max_{j\in A_\ell} e^{X_j}\right)_{1\le \ell\le d}. \tag{1.2}$$

Their continuous margins determine unique copulas denoted by $C_{\boldsymbol{S}}$ and $C_{\boldsymbol{Q}}$, respectively.
A first natural question is the following.

Q1. *How does the single-big-jump principle change when several group sums must be large simultaneously?*

Throughout the paper, vector inequalities and functions are interpreted componentwise and $\mathbf{1}$ denotes the vector of ones of the required dimension.
We show that under (1.1) there exists a finite constant $C_*\ge 1$ such that

$$\mathbb{P}\{\boldsymbol{S}>u\mathbf{1}\}\sim C_*\mathbb{P}\{\boldsymbol{Q}>u\mathbf{1}\},\qquad u\to\infty. \tag{1.3}$$

Theorem 2.1 gives sharp asymptotics for both probabilities and it identifies the constant C_*, which can be larger than one when $d>1$ as illustrated in Examples 3.1 and 3.2. Its value is determined by the dominant Gaussian configurations and the conditional contributions of the remaining risks in groups on the boundary of the exceedance region. Moreover, C_*^{-1} approximates the probability that all group maxima exceed u conditionally on simultaneous exceedance by the group sums. We obtain an exponential limit and mean asymptotics for the common excess above u. An extension to signed sums is established under the nondegeneracy conditions (3.13) in Proposition 3.5.
We also show that $\exp(\boldsymbol{X})$, $\boldsymbol{S}$ and $\boldsymbol{Q}$ are multivariate subexponential in the sense of Samorodnitsky and Sun [28]; the conclusion extends to both log-elliptical settings of Proposition 3.6. Multivariate subexponentiality does not imply $C_*=1$ since simultaneous upper orthants are not among the defining tail sets in dimension $d\ge 2$.
The dependence between the group totals raises a second question.

Q2. *What is the upper-tail dependence between group sums and how does it compare with that between group maxima?*

The distinction between asymptotic independence and residual dependence is central to this question. The dependence measures in [5, 31] quantify joint extremes beyond the usual upper tail-dependence coefficient. We show that $C_{\boldsymbol{S}}$ and $C_{\boldsymbol{Q}}$ are asymptotically independent and have the same joint upper-tail exponents. These exponents are determined by Gaussian quadratic optimisation problems. For each pair of groups, they yield an explicit residual dependence coefficient on the logarithmic scale. If the Gaussian variance is constant within each of the two groups this coefficient is $(1+\rho_*)/2$, where ρ_* is the largest correlation between Gaussian coordinates belonging to different groups. This extends the familiar Gaussian residual dependence formula [16] to group sums and maxima. For two groups with a common Gaussian variance across both groups, we also establish hidden regular variation after Pareto standardization, allowing arbitrary Gaussian means. When the upper tails of the group maxima are asymptotically equivalent across groups as in Remark 2.6, the constant C_* in (1.3) is also the limiting ratio of the joint upper-tail probabilities of the two copulas at common marginal probability levels. We also identify the extreme-value limit of the joint copula of $(\boldsymbol{S},\boldsymbol{Q})$.
Motivated by [14] we discuss similar questions for the multivariate extensions of signed sums in Proposition 3.5 below. Moreover, exploiting the radial representation of Gaussian random vectors we extend our findings to the log-elliptical setting.
By refining the exact tail asymptotic results obtained for both $\boldsymbol{S}$ and $\boldsymbol{Q}$ we discuss further several

applications including conditional approximations and their specialisation to conditional risk measures.

The paper is organised as follows. In Section 2 we present the sharp joint tail asymptotics and study the copulas of group sums and maxima. Section 3 is devoted to examples and extensions including common excess, signed sums, conditional risk measures and a log-elliptical extension with Gumbel radial tails. We shall relegate auxiliary results and proofs to Section 4.

2. Main Results

We consider the Gaussian setting (1.1) and the group sums and maxima defined in (1.2). For a nonempty index set $I \subset I_N := \{1, \dots, N\}$ write $\boldsymbol{x}_I$ for the subvector with indices in I, Σ_{II} for the corresponding principal submatrix and $\Sigma_{I_N,I}$ for the $N \times |I|$ matrix formed by the columns of Σ indexed by I; here $|B|$ denotes the cardinality of a finite set B. Indices are always listed in increasing order. Positive definiteness of Σ ensures that every Σ_{II} is invertible. Unless explicitly stated otherwise, group indices ℓ, r belong to $\{1, \dots, d\}$ and coordinate indices j, k belong to $\{1, \dots, N\}$.
We call a nonempty index set $I \subset I_N$ a selector if it contains at most one coordinate from each group, so that $|I \cap A_\ell| \le 1$ for every $1 \le \ell \le d$. For each selector define

$$\boldsymbol{\lambda}_I = \Sigma_{II}^{-1}\mathbf{1}_I, \qquad \boldsymbol{s}^I = \Sigma_{I_N,I}\boldsymbol{\lambda}_I, \qquad \kappa_I = \tfrac{1}{2}\mathbf{1}_I^{\mathsf{T}}\Sigma_{II}^{-1}\mathbf{1}_I. \tag{2.1}$$

Here $\boldsymbol{s}_I^I = \mathbf{1}_I$ and we denote further

$$\begin{aligned} &\mathfrak{I} = \{I : \boldsymbol{\lambda}_I > \mathbf{0}_I, \quad \max_{j \in A_\ell} s_j^I \ge 1 \text{ for every } 1 \le \ell \le d\}, \\ &\kappa = \min_{I \in \mathfrak{I}} \kappa_I, \qquad \mathfrak{I}_* = \{I \in \mathfrak{I} : \kappa_I = \kappa\}. \end{aligned} \tag{2.2}$$

The set $\mathfrak{I}$ is nonempty by Lemma 4.1 below and the selectors in $\mathfrak{I}$ are called admissible. The same lemma identifies $\boldsymbol{s}^I$, $I \in \mathfrak{I}_*$ as the configurations of minimum Gaussian quadratic cost subject to $\max_{j \in A_\ell} x_j \ge 1$ for every $1 \le \ell \le d$. For $I \in \mathfrak{I}_*$ set next

$$\begin{aligned} &B_I = \{1 \le \ell \le d : I \cap A_\ell \ne \varnothing\}, \quad \mathcal{C}_I = \{\ell \in \{1, \dots, d\} \setminus B_I : \max_{j \in A_\ell} s_j^I = 1\}, \\ &J_{\ell,I} = \{j \in A_\ell : s_j^I = 1\}, \qquad \ell \in \mathcal{C}_I \end{aligned} \tag{2.3}$$

and hence B_I contains the groups represented by a selected coordinate, while $\mathcal{C}_I$ contains the unrepresented groups whose largest coordinate in $\boldsymbol{s}^I$ equals one. In each such boundary group, $J_{\ell,I}$ is nonempty and collects the coordinates that contribute at scale u under the conditional limit below. The remaining unrepresented groups have $\max_{j \in A_\ell} s_j^I > 1$.
For any nonempty $I \subset \{1, \dots, N\}$, write $I^c = \{1, \dots, N\} \setminus I$. When $I^c \ne \varnothing$, we define

$$\begin{aligned} &D_I = \Sigma_{I^cI}\Sigma_{II}^{-1}, \qquad \Gamma_I = \Sigma_{I^cI^c} - D_I\Sigma_{II^c}, \\ &\boldsymbol{d}_I = \boldsymbol{\mu}_{I^c} - D_I\boldsymbol{\mu}_I, \quad \boldsymbol{Z}^{(I)} \sim \mathcal{N}(\mathbf{0}, \Gamma_I). \end{aligned} \tag{2.4}$$

The matrix Γ_I is positive definite and $\boldsymbol{d}_I + \boldsymbol{Z}^{(I)}$ has the same law as $\boldsymbol{X}_{I^c}|\boldsymbol{X}_I = \mathbf{0}_I$.
The contribution of the boundary groups is described by the random vectors $\boldsymbol{S}^{(I)}$ and $\boldsymbol{Q}^{(I)}$, indexed by $\mathcal{C}_I$, with joint law

$$\mathrm{Law}(\boldsymbol{S}^{(I)}, \boldsymbol{Q}^{(I)}) = \mathrm{Law}\left(\left(\sum_{j \in J_{\ell,I}} e^{X_j} \right)_{\ell \in \mathcal{C}_I}, \left(\max_{j \in J_{\ell,I}} e^{X_j} \right)_{\ell \in \mathcal{C}_I} \middle| \boldsymbol{X}_I = \mathbf{0}_I \right). \tag{2.5}$$

For every $I \in \mathfrak{I}_*$ we prove the following weak convergence (see the proof of Theorem 2.1)

$$\mathrm{Law}\left(\frac{\boldsymbol{S}_{\mathcal{C}_I}}{u}, \frac{\boldsymbol{Q}_{\mathcal{C}_I}}{u} \middle| e^{\boldsymbol{X}_I} > u\mathbf{1}_I \right) \xrightarrow[u \to \infty]{\mathrm{w}} \mathrm{Law}(\boldsymbol{S}^{(I)}, \boldsymbol{Q}^{(I)}). \tag{2.6}$$

The limiting margins have no atom at one implying that the corresponding conditional exceedance probabilities converge to $\mathbb{P}\{\boldsymbol{S}^{(I)} > \mathbf{1}\}$ and $\mathbb{P}\{\boldsymbol{Q}^{(I)} > \mathbf{1}\}$ as $u \to \infty$. If $\mathcal{C}_I = \varnothing$ these probabilities are set to 1.

Throughout, $\boldsymbol{T}=(T_\ell)_{1\le \ell\le d}\in\{\boldsymbol{S},\boldsymbol{Q}\}$ and $\boldsymbol{T}^{(I)}$ is the corresponding boundary vector in (2.5). The following theorem combines the dominant contributions.

Theorem 2.1. *As $u\to\infty$ we have*

$$\mathbb{P}\{\boldsymbol{T}>u\mathbf{1}\}\sim\sum_{I\in\mathfrak{I}_*}p_I(u)\mathbb{P}\{\boldsymbol{T}^{(I)}>\mathbf{1}\}, \tag{2.7}$$

where $p_I(u)=\mathbb{P}\{\boldsymbol{X}_I>(\log u)\mathbf{1}_I\}$, $I\in\mathfrak{I}_*$.

Remark 2.2. *Let $\mathfrak{H}$ consist of the sets $I\in\mathfrak{I}_*$ that maximize $\boldsymbol{\lambda}_I^\top\boldsymbol{\mu}_I$ and have minimum cardinality among these maximizers. The constant in (1.3) is given by*

$$C_*=\frac{\sum_{I\in\mathfrak{H}}A_I\mathbb{P}\{\boldsymbol{S}^{(I)}>\mathbf{1}\}}{\sum_{I\in\mathfrak{H}}A_I\mathbb{P}\{\boldsymbol{Q}^{(I)}>\mathbf{1}\}}\in[1,\infty), \tag{2.8}$$

where

$$A_I=\frac{\exp(-\boldsymbol{\mu}_I^\top\Sigma_{II}^{-1}\boldsymbol{\mu}_I/2)}{(2\pi)^{|I|/2}\sqrt{\det\Sigma_{II}}\prod_{j\in I}\lambda_{I,j}},\qquad I\in\mathfrak{I}_*.$$

If $d=1$, then $C_*=1$. For $d=2$ with common Gaussian variance, a singleton $I=\{j\}$ has $s_k^I=\Sigma_{kj}/\Sigma_{jj}<1$ for every $k\neq j$, by positive definiteness. It therefore cannot meet the constraint in the other group. Thus every $I\in\mathfrak{I}_*$ represents both groups and $\mathcal{C}_I=\varnothing$, giving $C_*=1$ also in this case. For $d\ge 3$, equal variances can still give $C_*>1$; see Example 3.2.

Next let $F_{\ell,\boldsymbol{T}}$ denote the distribution function of T_ℓ for $1\le\ell\le d$. The copula of $\boldsymbol{T}$ is

$$C_{\boldsymbol{T}}(\boldsymbol{v})=\mathbb{P}\Big\{T_\ell\le F_{\ell,\boldsymbol{T}}^{-1}(v_\ell),\ 1\le\ell\le d\Big\},\qquad \boldsymbol{v}\in(0,1)^d. \tag{2.9}$$

For comparing joint upper tails at a common marginal probability level $q\in(0,1)$ set below

$$J_{B,\boldsymbol{T}}(q)=\mathbb{P}\Big\{T_\ell>F_{\ell,\boldsymbol{T}}^{-1}(1-q),\ \ell\in B\Big\},\qquad \varnothing\neq B\subset\{1,\dots,d\} \tag{2.10}$$

and define $\sigma_\ell^*=\max_{j\in A_\ell}\sigma_j$, $1\le\ell\le d$; here $\sigma_j=\sqrt{\Sigma_{jj}}$, $\rho_{jk}=\Sigma_{jk}/(\sigma_j\sigma_k)$, $1\le j,k\le N$.

Proposition 2.3. *For every nonempty $B\subset\{1,\dots,d\}$*

$$J_{B,\boldsymbol{T}}(q)=q^{\beta_B+o(1)},\qquad q\downarrow 0, \tag{2.11}$$

where $\beta_{\{\ell\}}=1$, $1\le\ell\le d$ and

$$\beta_B=\min_{\boldsymbol{x}\in\mathbb{R}^N}\Big\{\boldsymbol{x}^\top\Sigma^{-1}\boldsymbol{x}:\max_{j\in A_\ell}x_j\ge\sigma_\ell^*,\ \ell\in B\Big\}. \tag{2.12}$$

If $|B|\ge 2$, then $\beta_B>1$ and if the Gaussian variances are constant within each of two distinct groups A_ℓ and A_r, then

$$\beta_{\{\ell,r\}}=\frac{2}{1+\max_{a\in A_\ell,\ b\in A_r}\rho_{ab}}. \tag{2.13}$$

For distinct ℓ,r, (2.11) gives the residual dependence coefficient on the logarithmic scale, namely

$$\eta_{\ell r,\boldsymbol{T}}:=\lim_{q\downarrow 0}\frac{\log q}{\log J_{\{\ell,r\},\boldsymbol{T}}(q)}=\frac{1}{\beta_{\{\ell,r\}}}.$$

When the Gaussian variances are constant within each group, (2.13) gives $\eta_{\ell r,\boldsymbol{T}}=(1+\max_{a\in A_\ell,\,b\in A_r}\rho_{ab})/2$. For singleton groups, this is the Gaussian residual dependence formula in [16, Section 2]; see for details on residual and weak tail dependence [5, 27, 31].

Our next finding concerns a sharp equivalent when $d\ge 2$ and the Gaussian margins are identical within each of the first two groups. Set $n_i=|A_i|$, $i=1,2$ and

$$\rho_*=\max_{a\in A_1,\ b\in A_2}\rho_{ab},\qquad n_*=\#\{(a,b)\in A_1\times A_2:\rho_{ab}=\rho_*\}.$$

Corollary 2.4. *For $i = 1, 2$ suppose $\mu_j = m_i$ and $\Sigma_{jj} = (\sigma_i^*)^2$ for $j \in A_i$, where $m_i \in \mathbb{R}$. If $B = \{1, 2\}$, then as $q \downarrow 0$*

$$J_{B,\boldsymbol{T}}(q) \sim K_* q^{2/(1+\rho_*)} [\log(1/q)]^{-\rho_*/(1+\rho_*)}, \tag{2.14}$$

where

$$K_* = \frac{n_*(1+\rho_*)^2}{\sqrt{1-\rho_*^2}} (n_1 n_2)^{-1/(1+\rho_*)} (4\pi)^{-\rho_*/(1+\rho_*)}. \tag{2.15}$$

For the case $d = 2$ below we define further $W_{i,\boldsymbol{T}} = (1 - F_{i,\boldsymbol{T}}(T_i))^{-1}$, $i = 1, 2$ and study the regular variation and the hidden regular variation; the case of Gaussian copulas has been already discussed in [7, 8].

Corollary 2.5. *If the Gaussian coordinates have common variance $\sigma^2 > 0$ and arbitrary means, then $(W_{1,\boldsymbol{T}}, W_{2,\boldsymbol{T}})$ is regularly varying with index one on $[0, \infty)^2 \setminus \{(0, 0)\}$ and has hidden regular variation on $(0, \infty)^2$ with index $\alpha_0 = 2/(1 + \rho_*)$.*

Remark 2.6. *If $\mathbb{P}\{Q_\ell > u\} \sim \mathbb{P}\{Q_1 > u\}$ as $u \to \infty$ for $1 \le \ell \le d$, then*

$$\frac{J_{B,\boldsymbol{S}}(q)}{J_{B,\boldsymbol{Q}}(q)} \longrightarrow C_*, \qquad q \downarrow 0, \quad B = \{1, \dots, d\}. \tag{2.16}$$

Let $C_{\boldsymbol{S},\boldsymbol{Q}}$ denote the copula of $(S_1, \dots, S_d, Q_1, \dots, Q_d)$.

Corollary 2.7. *Under (1.1), for $\boldsymbol{x}, \boldsymbol{y} \in [0, \infty)^d$*

$$\lim_{q \downarrow 0} \frac{1 - C_{\boldsymbol{S},\boldsymbol{Q}}(\boldsymbol{1} - q\boldsymbol{x}, \boldsymbol{1} - q\boldsymbol{y})}{q} = \sum_{\ell=1}^{d} \max(x_\ell, y_\ell). \tag{2.17}$$

Consequently, $C_{\boldsymbol{S},\boldsymbol{Q}}$ is in the max-domain of attraction of the extreme-value copula

$$C_{\mathrm{EV}}(\boldsymbol{u}, \boldsymbol{v}) = \prod_{\ell=1}^{d} \min(u_\ell, v_\ell), \qquad \boldsymbol{u}, \boldsymbol{v} \in [0, 1]^d. \tag{2.18}$$

The limit (2.17) describes exceedance unions at marginal probability scale q. Simultaneous exceedances of two or more groups have probability $o(q)$ by Proposition 2.3 and therefore their relative asymptotics are not determined by (2.18). In particular, the conditional probability in (3.1) can still converge to $C_*^{-1} < 1$, see Examples 3.1 and 3.2 below. The latter example has equal variances and $d = 3$. Complete upper-tail dependence within each individual pair therefore coexists with a nontrivial sum–maximum ratio under simultaneous conditioning.

3. Examples and Extensions

The constant C_* is strictly larger than one precisely when there exist $I \in \mathfrak{H}$ and $\ell \in \mathcal{C}_I$ with $|J_{\ell,I}| \ge 2$. Indeed, full support of the conditional Gaussian law then gives positive probability that this boundary sum exceeds one while its maximum is below one and all other boundary maxima exceed one; if every $J_{\ell,I}$ is a singleton, the boundary sums and maxima agree. Formula (2.8) proves the criterion, since all weights are positive. It also has the conditional interpretation

$$\mathbb{P}\{\boldsymbol{Q} > u\boldsymbol{1} \mid \boldsymbol{S} > u\boldsymbol{1}\} \longrightarrow C_*^{-1}, \qquad u \to \infty. \tag{3.1}$$

Example 3.1. *For independent standard normals Y_0, Y_1, Y_2 take $\boldsymbol{X} = (Y_0, Y_0 + Y_1, Y_0 + Y_2)^\mathsf{T}$, $A_1 = \{1\}$ and $A_2 = \{2, 3\}$ and hence*

$$\boldsymbol{S} = (e^{Y_0}, e^{Y_0+Y_1} + e^{Y_0+Y_2})^\mathsf{T} \tag{3.2}$$

and hence $C_* = \frac{4}{3}\mathbb{P}\Big\{e^{Y_1} + e^{Y_2} > 1\Big\} \in (1, 4/3)$ with numerical value $C_* \approx 1.182066$.

Example 3.2. *Let G_0, G_1, G_2 and $(\varepsilon_{i,k})_{1\le i\le 2, k\in\mathbb{Z}}$ be independent standard normals and set*

$$X_i(k) = \frac{G_0}{2} + \frac{G_1\cos(k\pi/3) + G_2\sin(k\pi/3)}{\sqrt{2}} + \frac{\varepsilon_{i,k}}{2}, \qquad i\in\{1,2\}, \quad k\in\mathbb{Z}. \tag{3.3}$$

This stationary Gaussian process has unit variances and positive definite covariances for finite samples of distinct coordinate–time pairs. For the three groups $S_k = e^{X_1(k)}+e^{X_2(k)}$, $k=0,1,2$, we have $1<C_*<3/2$, with numerical value $C_*\approx 1.332802$. Their identical marginal laws also give the copula ratio (2.16). For every pair of these groups, $C_*=1$ and the two aggregate copulas have the same asymptotics in (2.14).

Remark 3.3. *Let $(X_1(k), X_2(k))$, $k\in\mathbb{Z}$ be stationary Gaussian with means μ_i, $i=1,2$ and common component variance $v>0$. On finite nonempty grids $\mathcal{T}_i\subset\mathbb{Z}$, put $S_i=\sum_{k\in\mathcal{T}_i} e^{X_i(k)}$ and $Q_i=\max_{k\in\mathcal{T}_i} e^{X_i(k)}$, $i=1,2$ and let*

$$c_* = \max_{(k,l)\in\mathcal{T}_1\times\mathcal{T}_2} \mathrm{Cov}(X_1(k), X_2(l)). \tag{3.4}$$

Assume that $(X_1(k), k\in\mathcal{T}_1;\ X_2(l), l\in\mathcal{T}_2)$ has a positive definite covariance matrix. If n_* counts the pairs attaining c_*, then for one such pair (k_*, l_*)

$$\mathbb{P}\{\boldsymbol{S}>u\mathbf{1}\}\sim\mathbb{P}\{\boldsymbol{Q}>u\mathbf{1}\}\sim n_*\mathbb{P}\{X_1(k_*)>\log u,\ X_2(l_*)>\log u\}, \qquad u\to\infty. \tag{3.5}$$

The copula equivalent in Corollary 2.4 holds with $n_i=|\mathcal{T}_i|$, $i\in\{1,2\}$, $\rho_*=c_*/v$ and this n_*.

More generally, a finite Gaussian sample may be grouped by component or by observation time. Theorem 2.1 applies whenever its covariance is positive definite; stationarity is not required.

Example 3.4. *Let $N\ge 2$, $\boldsymbol{\mu}=m\mathbf{1}$ and*

$$\Sigma = \sigma^2[(1-\rho)\mathrm{Id}_N + \rho\mathbf{1}\mathbf{1}^{\mathsf{T}}], \qquad \sigma>0, \quad -\frac{1}{N-1}<\rho<1,$$

where Id_N is the identity matrix. Put $n_\ell=|A_\ell|$ and $D_k=1+(k-1)\rho$, $1\le k\le N$. It follows that

$$\kappa=\frac{d}{2\sigma^2 D_d}, \qquad C_*=1.$$

Writing $z_u=(\log u-m)/\sigma$ we have as $u\to\infty$

$$\mathbb{P}\{\boldsymbol{S}>u\mathbf{1}\}\sim\mathbb{P}\{\boldsymbol{Q}>u\mathbf{1}\}\sim\left(\prod_{\ell=1}^{d} n_\ell\right)\frac{D_d^{d-1/2}}{(2\pi)^{d/2}(1-\rho)^{(d-1)/2}} z_u^{-d}\exp\left(-\frac{dz_u^2}{2D_d}\right) \tag{3.6}$$

and for any nonempty $B\subset\{1,\dots,d\}$ with $b=|B|$

$$\beta_B=\frac{b}{1+(b-1)\rho}. \tag{3.7}$$

Consequently, the joint extreme-value copula (2.18) is the same for every admissible ρ, while the simultaneous upper-tail exponents vary with ρ.

3.1. **Aggregation and common excess given exceedance.** We return to the positive group sums and maxima in (1.2). We write $\mathrm{Exp}(\lambda)$ for the exponential law with rate $\lambda>0$. Define $S_\wedge=\min_{1\le\ell\le d} S_\ell$ and $R_u=S_\wedge/\min_{1\le\ell\le d} Q_\ell$ and hence conditionally on $\boldsymbol{S}>u\mathbf{1}$

$$R_u \xrightarrow[u\to\infty]{\mathrm{w}} \mathcal{R}, \qquad \mathbb{P}\{\mathcal{R}=1\}=C_*^{-1}. \tag{3.8}$$

Consequently, we have that $C_*=1$ iff $R_u\xrightarrow[u\to\infty]{\mathbb{P}}1$ under this conditioning. Moreover, we have

$$\mathbb{P}\{\mathcal{R}\le r\}=\frac{\sum_{I\in\mathfrak{H}} A_I\mathbb{P}\{\boldsymbol{S}^{(I)}>\mathbf{1},\ \boldsymbol{Q}^{(I)}>r^{-1}\mathbf{1}\}}{\sum_{I\in\mathfrak{H}} A_I\mathbb{P}\{\boldsymbol{S}^{(I)}>\mathbf{1}\}}, \qquad r\ge 1, \tag{3.9}$$

with $\mathbb{P}\{\mathcal{R} \le r\} = 0$ for $r < 1$. Furthermore, as $u \to \infty$

$$\text{Law}\left(\frac{2\kappa \log u}{u}(S_\wedge - u)\middle| \boldsymbol{S} > u\mathbf{1}\right) \xrightarrow[u\to\infty]{\text{w}} \text{Exp}(1), \quad \mathbb{E}\{S_\wedge - u \mid \boldsymbol{S} > u\mathbf{1}\} \sim \frac{u}{2\kappa \log u}. \tag{3.10}$$

For applications to basket options and options on minima, see e.g., [10, 30].

3.2. **Signed sums.** With motivation from [14] that considers the case $d = 1$ we discuss briefly the case of signed sums. For $1 \le \ell \le d$ split $A_\ell = P_\ell \dot\cup N_\ell$ with $P_\ell \ne \varnothing$ and define

$$\widetilde{S}_\ell = \sum_{j\in P_\ell} e^{X_j} - \sum_{k\in N_\ell} e^{X_k}, \qquad \widetilde{Q}_\ell \ = \max_{j\in P_\ell} e^{X_j} - \sum_{k\in N_\ell} e^{X_k} \tag{3.11}$$

setting further $\widetilde{\boldsymbol{S}} = (\widetilde{S}_\ell)_{1\le\ell\le d}$ and $\widetilde{\boldsymbol{Q}} = (\widetilde{Q}_\ell)_{1\le\ell\le d}$. Nonzero deterministic weights can be absorbed into the means and signs. With $\max\varnothing = -\infty$, put

$$\mathcal{F}_\pm = \left\{\boldsymbol{x}\in\mathbb{R}^N : \max_{j\in P_\ell} x_j \ge \max\left(1, \max_{k\in N_\ell} x_k\right),\ 1\le\ell\le d\right\}, \qquad \mathcal{S}_\pm \ = \operatorname*{argmin}_{\boldsymbol{x}\in\mathcal{F}_\pm} \tfrac{1}{2}\boldsymbol{x}^\mathsf{T}\Sigma^{-1}\boldsymbol{x}. \tag{3.12}$$

For each $\boldsymbol{s}\in\mathcal{S}_\pm$, write $\boldsymbol{g} = \Sigma^{-1}\boldsymbol{s}$ and $h_\ell = \max_{j\in P_\ell} s_j$, $1\le\ell\le d$. The nondegeneracy assumption is that for every $1\le\ell\le d$ with $\boldsymbol{g}_{A_\ell} \ne \mathbf{0}$

$$h_\ell = 1 \implies \sum_{j\in A_\ell} g_j > 0, \qquad s_k = h_\ell \implies g_k < 0, \quad k\in N_\ell. \tag{3.13}$$

Proposition 3.5. *Under* (3.13) *for some* $C_\pm \in [1,\infty)$

$$\mathbb{P}\left\{\widetilde{\boldsymbol{S}} > u\mathbf{1}\right\} \sim C_\pm \mathbb{P}\left\{\widetilde{\boldsymbol{Q}} > u\mathbf{1}\right\}, \qquad u\to\infty, \tag{3.14}$$

where $C_\pm = 1$ *when* $d = 1$.

Without negative terms this reduces to (1.3). For the scalar case, see also [14, Theorem 2] under its nondegeneracy assumptions. Under Proposition 3.5, the common-excess conclusions (3.10) also hold with $S_\wedge$ replaced by $\min_{1\le\ell\le d}\widetilde{S}_\ell$, conditioning on $\widetilde{\boldsymbol{S}} > u\mathbf{1}$ and with $\kappa_\pm = \min_{\boldsymbol{x}\in\mathcal{F}_\pm}\frac{1}{2}\boldsymbol{x}^\mathsf{T}\Sigma^{-1}\boldsymbol{x}$ in place of κ.

3.3. **Conditional risk measures.** Use the Gaussian sums in (1.1)–(1.2); for related conditional risk measures, see e.g., [20]. For $\varnothing \ne B \subset \{1,\dots,d\}$ and $u > 0$, put $\mathcal{E}_B(u) = \{S_r > u,\ r\in B\}$. Write $\mathbb{P}_{\boldsymbol{\nu}}$ for probability under $\mathcal{N}_N(\boldsymbol{\nu},\Sigma)$, $\boldsymbol{\nu}\in\mathbb{R}^N$ and define $\boldsymbol{\mu}^{(j)} = \boldsymbol{\mu} + \Sigma\boldsymbol{e}_j$, $1\le j\le N$, where $\boldsymbol{e}_j, 1\le j\le N$ denotes the jth coordinate vector. Gaussian tilting implies for $1\le\ell\le d$ (see e.g., [19, Lem 6.1])

$$\mathbb{E}\{S_\ell \mid \mathcal{E}_B(u)\} = \sum_{j\in A_\ell} e^{\mu_j + \Sigma_{jj}/2}\frac{\mathbb{P}_{\boldsymbol{\mu}^{(j)}}\{\mathcal{E}_B(u)\}}{\mathbb{P}_{\boldsymbol{\mu}}\{\mathcal{E}_B(u)\}}. \tag{3.15}$$

In view of Theorem 2.1 we have a sharp equivalent by evaluating the tails for the groups in B at each corresponding mean. For $B = \{1,\dots,d\}$ this implies the conditional tail expectation; for $\ell\notin B$, the marginal expected shortfall under stress in B.

Under Remark 3.3, setting $n_i = |\mathcal{T}_i|$, $i = 1,2$, for $i,j\in\{1,2\}$ with $i\ne j$ and $u > 0$ define the marginal expected shortfall and marginal mean excess by

$$\text{MES}_i(u) = \mathbb{E}\{S_i \mid S_j > u\}, \qquad \text{MME}_i(u) = \mathbb{E}\{(S_i - u)_+ \mid S_j > u\},$$

where $x_+ = \max(x,0)$. For these measures under regular variation and asymptotic tail independence, see [6]. For truncated moments, including those of multivariate lognormal vectors, see e.g., [1].

Next as $u\to\infty$ we have ($\overline{\Phi}$ denotes the standard normal survival function)

$$\text{MES}_i(u) \sim \frac{n_*}{n_j}\exp\left(\mu_i + \frac{v}{2} - \frac{c_*\mu_j}{v} - \frac{c_*^2}{2v}\right)u^{c_*/v}, \tag{3.16}$$

$$\text{MME}_i(u) \sim (v + c_*)\frac{u}{\log u}\frac{\mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\}}{\mathbb{P}\{S_j > u\}}, \tag{3.17}$$

where the tail ratio in (3.17) follows from (3.5) and $\mathbb{P}\{S_j > u\} \sim n_j\overline{\Phi}((\log u - \mu_j)/\sqrt{v})$.

3.4. **Log-elliptical sums.** The tractability of lognormal random vectors strongly relates to the radial decomposition of the Gaussian distribution, which is shared also by elliptical random vectors. Therefore in this subsection and its proof we shall replace (1.1) by

$$\boldsymbol{X} = \boldsymbol{\mu} + RA\boldsymbol{U}_N, \qquad \Sigma = AA^{\mathsf{T}} \succ 0, \qquad R \perp \boldsymbol{U}_N, \qquad \boldsymbol{U}_N \sim \mathrm{Unif}(\mathbb{S}^{N-1}), \tag{3.18}$$

where $A \in \mathbb{R}^{N\times N}$ and $\mathbb{S}^{N-1}$ is the unit sphere in $\mathbb{R}^N$. Here $\Sigma = AA^{\mathsf{T}}$ is the scatter matrix of $\boldsymbol{X}$. Use the positive aggregates (1.2) and the selector notation in (2.1)–(2.3) and set $p_I(u) = \mathbb{P}\Big\{e^{\boldsymbol{X}_I} > u\mathbf{1}_I\Big\}$, $I \in \mathfrak{I}_*$, under (3.18). Suppose $R > 0$ has infinite upper endpoint and survival function $\overline{F}$ in the Gumbel max-domain of attraction: for a positive measurable scaling function w

$$\lim_{u\to\infty} \frac{\overline{F}(u + x/w(u))}{\overline{F}(u)} = e^{-x}, \qquad x \in \mathbb{R}. \tag{3.19}$$

Since the upper endpoint is infinite, this condition implies $\lim_{\xi\to\infty} \xi w(\xi) = \infty$; see [15, Remark 3.2(b), Eq. (3.16)]. We shall write below $w \in \mathrm{RV}_a$ if $\lim_{u\to\infty} w(cu)/w(u) = c^a$ for every fixed $c > 0$, i.e., w is regularly varying at infinity with index a.

For $\tau > 0$ define $\boldsymbol{S}_\tau^{(I)}, \boldsymbol{Q}_\tau^{(I)}$ by (2.5) using the Gaussian reference law $\mathcal{N}_N(\boldsymbol{\mu}, \tau^2\Sigma)$. Also let $\boldsymbol{Z}^{(I)} \sim \mathcal{N}_{N-|I|}(\mathbf{0}, \Gamma_I)$ for $I \in \mathfrak{I}_*$ with $I^c \neq \varnothing$, where Γ_I is as in (2.4). Boundary probabilities are one when $\mathcal{C}_I = \varnothing$.

Proposition 3.6. (i) *If* $\lim_{u\to\infty} u/w(u) = \tau^2 \in (0,\infty)$, *then*

$$\mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\} \sim \sum_{I\in\mathfrak{I}_*} p_I(u)\mathbb{P}\Big\{\boldsymbol{T}_\tau^{(I)} > \mathbf{1}\Big\}, \qquad u \to \infty. \tag{3.20}$$

Moreover, $\lim_{u\to\infty} \mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\}/\mathbb{P}\{\boldsymbol{Q} > u\mathbf{1}\} = C_\tau$, *where* $C_\tau \in [1,\infty)$ *is the constant* (2.8) *for the Gaussian reference law* $\mathcal{N}_N(\boldsymbol{\mu}, \tau^2\Sigma)$.

(ii) *If* $w \in \mathrm{RV}_a$ *with* $-1 < a < 1$, *then*

$$\mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\} \sim \mathbb{P}\{\boldsymbol{Q} > u\mathbf{1}\} \sim \sum_{I\in\mathfrak{I}_*} p_I(u)\mathbb{P}\Big\{\max_{j\in J_{\ell,I}} Z_j^{(I)} > 0,\ \ell \in \mathcal{C}_I\Big\}, \qquad u \to \infty. \tag{3.21}$$

Under either part of Proposition 3.6, with $a = 1$ in part (i), the grouped copulas have for $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$ and $1 \le \ell < r \le d$

$$\eta_{\ell r,\boldsymbol{T}} = \beta_{\{\ell,r\}}^{-(a+1)/2}, \qquad \chi_{\ell r,\boldsymbol{T}} = 0, \qquad \bar{\chi}_{\ell r,\boldsymbol{T}} = 2\eta_{\ell r,\boldsymbol{T}} - 1. \tag{3.22}$$

Here the dependence coefficients are defined by

$$\chi_{\ell r,\boldsymbol{T}} := \lim_{q\downarrow 0} \frac{J_{\{\ell,r\},\boldsymbol{T}}(q)}{q}, \quad \bar{\chi}_{\ell r,\boldsymbol{T}} := \lim_{q\downarrow 0} \frac{2\log q}{\log J_{\{\ell,r\},\boldsymbol{T}}(q)} - 1.$$

The exponent β_B is given by (2.12) with the scatter matrix Σ. Thus both grouped copulas remain asymptotically independent. If the scatter diagonal is constant within each of A_ℓ, A_r, then (2.13) gives

$$\eta_{\ell r,\boldsymbol{T}} = \left(\frac{1+\rho^*_{\ell r}}{2}\right)^{(a+1)/2}, \qquad \rho^*_{\ell r} = \max_{j\in A_\ell,\ k\in A_r} \frac{\Sigma_{jk}}{\sqrt{\Sigma_{jj}\Sigma_{kk}}}.$$

For singleton groups this recovers [16, Theorem 2.1].

A further direction is to consider log-Liouville models, with $\boldsymbol{U}_N$ a Dirichlet random vector $\boldsymbol{D}$ with positive parameters on the positive L^1 unit sphere $\{\boldsymbol{x} \in [0,\infty)^N : \sum_{j=1}^N x_j = 1\}$. The radial representation suggests a similar radial analysis, with the simplex geometry requiring separate treatment of the boundary terms; for extremal attractors of Liouville copulas and their survival counterparts see [4].

3.5. **Subexponentiality.** Subexponentiality for random vectors has been discussed in [28]. For an integer $m \geq 1$ let $\mathfrak{R}_m$ consist of the nonempty open increasing sets $G \subset \mathbb{R}^m$ with convex complements and $\mathbf{0} \notin \overline{G}$. A nonnegative random vector $\boldsymbol{V}$ belongs to $\mathcal{S}_{\mathfrak{R}_m}$ if $u \mapsto \mathbb{P}\{\boldsymbol{V} \in uG\}$ is a univariate subexponential tail for every $G \in \mathfrak{R}_m$; see [22, 28].
Under (1.1) or either part of Proposition 3.6, $\exp(\boldsymbol{X}) \in \mathcal{S}_{\mathfrak{R}_N}$ and $\boldsymbol{S}, \boldsymbol{Q} \in \mathcal{S}_{\mathfrak{R}_d}$. For $d \geq 2$ however $(1,\infty)^d \notin \mathfrak{R}_d$ since its complement is not convex; see [22, Remark 2.2]. These membership criteria do not determine C_*; in particular the Gaussian cases with $C_* > 1$ remain in this class.
The proof in the next section verifies [28, Definition 4.6] directly, using its Lemma 4.5. Their Proposition 4.15 does not cover the whole log-elliptical regime: its dominated-variation assumption fails for $0 < a < 1$. The corresponding convolution results concern sums of independent vector copies. For simultaneous tail events, several large jumps may contribute; see [9].

4. Further Results and Proofs

We next present some auxiliary results, writing $t_u = \log u$ and $E_u = \{\boldsymbol{S} > u\mathbf{1}\}$. For $t \in \mathbb{R}, I \in \mathfrak{I}_*$ set

$$\boldsymbol{z}_I(t) = t\mathbf{1}_I - \boldsymbol{\mu}_I, \qquad \boldsymbol{g}_I(t) = \Sigma_{II}^{-1}\boldsymbol{z}_I(t), \qquad r_I(t) = (\boldsymbol{z}_I(t)^\mathsf{T}\Sigma_{II}^{-1}\boldsymbol{z}_I(t))^{1/2}. \tag{4.1}$$

For any nonempty $I \subset I_N$ we write (recall D_I in (2.4))

$$\boldsymbol{\Delta}_I = \boldsymbol{X}_I - t_u\mathbf{1}_I, \qquad \boldsymbol{\eta}_I = \boldsymbol{X}_{I^c} - \boldsymbol{\mu}_{I^c} - D_I(\boldsymbol{X}_I - \boldsymbol{\mu}_I). \tag{4.2}$$

The residual $\boldsymbol{\eta}_I$ is independent of $\boldsymbol{X}_I$ with law $\mathcal{N}(\mathbf{0}, \Gamma_I)$; empty residuals are omitted. Set next

$$\Omega = \Sigma^{-1}, \qquad q(\boldsymbol{x}) = \tfrac{1}{2}\boldsymbol{x}^\mathsf{T}\Omega\boldsymbol{x}, \qquad \mathcal{F} = \{\boldsymbol{x} \in \mathbb{R}^N : \max_{j \in A_\ell} x_j \geq 1, 1 \leq \ell \leq d\}. \tag{4.3}$$

Lemma 4.1.

$$\mathfrak{I} \neq \varnothing, \qquad \min_{\boldsymbol{x} \in \mathcal{F}} q(\boldsymbol{x}) = \kappa, \qquad \operatorname*{argmin}_{\boldsymbol{x} \in \mathcal{F}} q(\boldsymbol{x}) = \{\boldsymbol{s}^I : I \in \mathfrak{I}_*\}.$$

Moreover, we have

$$s_k^I < 1, \qquad I \in \mathfrak{I}_*, \quad \ell \in B_I, \quad k \in A_\ell \setminus I. \tag{4.4}$$

For the following asymptotic equivalence and lemma, let $\varnothing \neq I \subset \{1, \dots, N\}$ satisfy $\boldsymbol{\lambda}_I = \Sigma_{II}^{-1}\mathbf{1}_I > \mathbf{0}_I$. For bounded $\boldsymbol{\beta}_I(t)$, put

$$\boldsymbol{z}_I(t) = t\mathbf{1}_I + \boldsymbol{\beta}_I(t), \qquad \boldsymbol{g}_I(t) = \Sigma_{II}^{-1}\boldsymbol{z}_I(t), \qquad r_I(t)^2 = \boldsymbol{z}_I(t)^\mathsf{T}\boldsymbol{g}_I(t).$$

The choice $\boldsymbol{\beta}_I(t) = -\boldsymbol{\mu}_I$ recovers (4.1). The Gaussian orthant equivalent [21] is

$$\mathbb{P}\{\boldsymbol{X}_I > \boldsymbol{\mu}_I + \boldsymbol{z}_I(t)\} \sim \frac{e^{-r_I(t)^2/2}}{(2\pi)^{|I|/2}\sqrt{\det \Sigma_{II}}\prod_{j \in I} g_{I,j}(t)}, \qquad t \to \infty. \tag{4.5}$$

Lemma 4.2. *With the preceding notation and the residuals in (4.2)*

$$\operatorname{Law}\left(t(\boldsymbol{X}_I - \boldsymbol{\mu}_I - \boldsymbol{z}_I(t)), \boldsymbol{\eta}_I \mid \boldsymbol{X}_I > \boldsymbol{\mu}_I + \boldsymbol{z}_I(t)\right) \xrightarrow[t\to\infty]{\mathrm{w}} \left(\bigotimes_{j \in I} \operatorname{Exp}(\lambda_{I,j})\right) \otimes \mathcal{N}_{N-|I|}(\mathbf{0}, \Gamma_I). \tag{4.6}$$

If $I^c = \varnothing$, then we omit the residuals and Gaussian factor.

Proof of Lemma 4.1. The set $\mathcal{F}$ in (4.3) is the finite union of $\{\boldsymbol{x} : \boldsymbol{x}_K \geq \mathbf{1}_K\}$ over sets K selecting one coordinate from each group. Minimizing first over $\boldsymbol{x}_{K^c}$ reduces each problem to

$$\min_{\boldsymbol{y} \geq \mathbf{1}_K} \tfrac{1}{2}\boldsymbol{y}^\mathsf{T}\Sigma_{KK}^{-1}\boldsymbol{y}.$$

By [20, Lemma 1, Eqs. (20)–(22)], its unique minimizer is $\boldsymbol{y} = \Sigma_{K,I}\boldsymbol{\lambda}_I$ for some nonempty $I \subset K$ with $\boldsymbol{\lambda}_I > \mathbf{0}_I$. The minimizing extension $\Sigma_{I_N,K}\Sigma_{KK}^{-1}\boldsymbol{y}$ is therefore $\boldsymbol{s}^I$ and has cost κ_I. Thus $I \in \mathfrak{I}$. Conversely, every $I \in \mathfrak{I}$ gives a feasible point $\boldsymbol{s}^I$ with cost κ_I. Taking the minimum over the finite union proves the first three assertions.
Finally, if $I \in \mathfrak{I}_*$ and another coordinate $k \neq j$ in the group of $j \in I$ satisfied $s_k^I \geq 1$, then $\boldsymbol{s}^I - \varepsilon\boldsymbol{e}_j$ would remain feasible, whereas

$$q(\boldsymbol{s}^I - \varepsilon\boldsymbol{e}_j) = \kappa - \varepsilon\lambda_{I,j} + \tfrac{1}{2}\varepsilon^2\Omega_{jj} < \kappa$$

for small $\varepsilon>0$. This proves (4.4). □

Proof of (4.5) and Lemma 4.2. Apply [20, Lemma 2, Eq. (25)] to $\boldsymbol{X}_I-\boldsymbol{\mu}_I$ with covariance Σ_{II}, direction $\mathbf{1}_I$ and perturbation $\boldsymbol{\beta}_I(t)=o(t)$. All coordinates are active since $\Sigma_{II}^{-1}\mathbf{1}_I=\boldsymbol{\lambda}_I>\mathbf{0}_I$. Since $g_{I,j}(t)\sim t\lambda_{I,j}$ for $j\in I$, the cited tail equivalent gives (4.5). Applying it also at the shifted threshold yields, for every $\boldsymbol{v}\geq\mathbf{0}_I$,

$$\lim_{t\to\infty}\frac{\mathbb{P}\{\boldsymbol{X}_I>\boldsymbol{\mu}_I+\boldsymbol{z}_I(t)+\boldsymbol{v}/t\}}{\mathbb{P}\{\boldsymbol{X}_I>\boldsymbol{\mu}_I+\boldsymbol{z}_I(t)\}}=e^{-\boldsymbol{\lambda}_I^{\top}\boldsymbol{v}}. \tag{4.7}$$

This proves the exponential-vector limit. Gaussian regression makes $\boldsymbol{\eta}_I$ independent of $\boldsymbol{X}_I$ with law $\mathcal{N}(\mathbf{0},\Gamma_I)$, and hence the joint convergence (4.6) follows. Empty residuals are omitted. □

Proof of Theorem 2.1. Let $u\to\infty$ and put $t_u=\log u$. For $I\in\mathfrak{I}_*$, let $H_I(u)=\{\boldsymbol{X}_I>t_u\mathbf{1}_I\}$. In view of Lemma 4.2,conditionally on $H_I(u)$

$$\boldsymbol{\Delta}_I=O_{\mathbb{P}}(t_u^{-1}),\qquad \boldsymbol{X}/t_u\xrightarrow[u\to\infty]{\mathbb{P}}\boldsymbol{s}^I. \tag{4.8}$$

The orthant equivalent (4.5) also gives, as $u\to\infty$, the threshold stability

$$\mathbb{P}\{\boldsymbol{X}_I>t_u\mathbf{1}_I-\boldsymbol{\epsilon}(t_u)\}\sim p_I(u)\quad\text{if }\|\boldsymbol{\epsilon}(t_u)\|=o(t_u^{-1}). \tag{4.9}$$

Use q and $\mathcal{F}$ from (4.3). For $I\in\mathfrak{I}_*$, $\Sigma^{-1}\boldsymbol{s}^I$ is positive exactly on I and zero elsewhere; hence distinct sets I give distinct minimizers. By Lemma 4.1, these minimizers have disjoint open neighbourhoods U_I, $I\in\mathfrak{I}_*$, on which $x_j\leq 1-\delta$ whenever $j\in A_\ell\setminus I$ and $\ell\in B_I$, for some $\delta>0$. Put $U=\bigcup_{I\in\mathfrak{I}_*}U_I$. For sufficiently small $\varepsilon>0$, coercivity gives $\vartheta>0$ such that

$$\inf\{q(\boldsymbol{x}):\boldsymbol{x}\notin U,\max_{j\in A_\ell}x_j\geq 1-\varepsilon,\ 1\leq\ell\leq d\}\geq\kappa+2\vartheta. \tag{4.10}$$

Otherwise a bounded sequence would converge to a minimizer outside U. On E_u, $\max_{j\in A_\ell}X_j>t_u-\log|A_\ell|$ for every $1\leq\ell\leq d$. By (4.10), on $E_u\cap\{\boldsymbol{X}/t_u\notin U\}$ and for large u,

$$\|\Sigma^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})\|\geq t_u\sqrt{2(\kappa+2\vartheta)}-\|\Sigma^{-1/2}\boldsymbol{\mu}\|\geq t_u\sqrt{2(\kappa+3\vartheta/2)}.$$

Since $\|\Sigma^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})\|^2$ has a χ^2_N distribution, its tail bound, with the polynomial factor absorbed into the remaining exponential gap, yields

$$\mathbb{P}\{E_u,\boldsymbol{X}/t_u\notin U\}\leq e^{-(\kappa+\vartheta)t_u^2}=o(p_I(u)),\qquad I\in\mathfrak{I}_*; \tag{4.11}$$

here (4.5) implies $\log p_I(u)=-\kappa t_u^2+O(t_u+\log t_u)$.
Fix $I\in\mathfrak{I}_*$. On $\{\boldsymbol{X}/t_u\in U_I\}$, a represented group has one selected term and all other terms are at most $ue^{-\delta t_u}$. For $j\in I\cap A_\ell$, set

$$\epsilon_j(t_u)=-\log(1-(|A_\ell|-1)e^{-\delta t_u})=O(e^{-\delta t_u})$$

and $H_I^-=\{\boldsymbol{X}_I>t_u\mathbf{1}_I-\boldsymbol{\epsilon}(t_u)\}$. The represented-group sum conditions lie between H_I and H_I^- and (4.9) gives

$$\mathbb{P}\Big\{H_I^-\setminus H_I\Big\}=o(p_I(u)). \tag{4.12}$$

Together with (4.8), this yields

$$\mathbb{P}\{E_u,\boldsymbol{X}/t_u\in U_I\}=p_I(u)\mathbb{P}\{S_\ell>u,\ \ell\in\{1,\dots,d\}\setminus B_I\mid H_I(u)\}+o(p_I(u)). \tag{4.13}$$

Now, by Gaussian regression

$$(e^{X_j}/u)_{j\in I^c}=\exp\{t_u(\boldsymbol{s}^I_{I^c}-\mathbf{1}_{I^c})+\boldsymbol{d}_I+D_I\boldsymbol{\Delta}_I+\boldsymbol{\eta}_I\}. \tag{4.14}$$

and in view of (4.6) the unrepresented groups with $\max_{j\in A_\ell}s^I_j>1$ exceed u with conditional probability tending to one, while the boundary groups have the joint limit (2.6). Since $\Gamma_I\succ 0$, the relevant residual coordinates have a Lebesgue density. The level sets of their exponential sums and maxima at one are Lebesgue-null, so the limiting margins have no atom at one. Hence the conditional probability in (4.13) converges to $\mathbb{P}\{\boldsymbol{S}^{(I)}>\mathbf{1}\}>0$. Summing over I and using (4.11) proves the sum equivalent.

For maxima, the represented-group conditions on U_I are exactly H_I for large u. The same argument, with maxima in (4.13), proves (2.7) for $\boldsymbol{T} = \boldsymbol{Q}$. Full support of the residual Gaussian law gives $0 < \mathbb{P}\{\boldsymbol{Q}^{(I)} > \mathbf{1}\} \le \mathbb{P}\{\boldsymbol{S}^{(I)} > \mathbf{1}\} \le 1$. Finally, the finite positive sums in (2.7) give

$$\lim_{u\to\infty} \frac{\log \mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\}}{(\log u)^2} = -\kappa, \qquad \boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}. \tag{4.15}$$

Expanding the exponent in (4.5) gives, as $u \to \infty$ with $t_u = \log u$,

$$p_I(u) \sim A_I t_u^{-|I|} e^{-\kappa t_u^2 + (\boldsymbol{\lambda}_I^\mathsf{T}\boldsymbol{\mu}_I)t_u}, \qquad I \in \mathfrak{I}_*. \tag{4.16}$$

For any $I_0 \in \mathfrak{H}$ the definition of $\mathfrak{H}$ therefore implies $\lim_{u\to\infty} p_I(u)/p_{I_0}(u) = (A_I/A_{I_0})\mathbb{I}_{\{I\in\mathfrak{H}\}}$, where $\mathbb{I}_{\{\cdot\}}$ is the indicator function. Consequently, (2.7) yields

$$\mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\} \sim \frac{p_{I_0}(u)}{A_{I_0}} \sum_{I\in\mathfrak{H}} A_I \mathbb{P}\Big\{\boldsymbol{T}^{(I)} > \mathbf{1}\Big\}, \qquad \boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}. \tag{4.17}$$

All weights and boundary probabilities are positive and $\mathbb{P}\{\boldsymbol{Q}^{(I)} > \mathbf{1}\} \le \mathbb{P}\{\boldsymbol{S}^{(I)} > \mathbf{1}\}$ for every $I \in \mathfrak{H}$. Taking the ratio in (4.17) proves (1.3) with the constant (2.8), which belongs to $[1, \infty)$. □

Proof of Proposition 2.3. Fix $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$ and let $q \downarrow 0$. Put $z_q = \overline{\Phi}^{-1}(q)$, so $\lim_{q\downarrow 0} z_q = \infty$ and $z_q \sim \sqrt{2\log(1/q)}$ as $q \downarrow 0$. The bounds $Q_\ell \le T_\ell \le |A_\ell| Q_\ell$, together with the union bounds for the Gaussian maximum, give

$$\lim_{x\to\infty} x^{-2} \log \mathbb{P}\{\log T_\ell > x\} = -\frac{1}{2(\sigma_\ell^*)^2}, \qquad \lim_{q\downarrow 0} \frac{\log F^{-1}_{\ell,\boldsymbol{T}}(1-q)}{z_q} = \sigma_\ell^*.$$

For thresholds $h_\ell(z) = z\sigma_\ell^* + o(z)$, $\ell \in B$, we claim that

$$\lim_{z\to\infty} \frac{2}{z^2} \log \mathbb{P}\Big\{\max_{j\in A_\ell} X_j > h_\ell(z),\ \ell \in B\Big\} = -\beta_B.$$

Indeed, for every small $\varepsilon > 0$, the normalized thresholds are eventually between $(1-\varepsilon)\sigma_\ell^*$ and $(1+\varepsilon)\sigma_\ell^*$. By homogeneity, the minimum quadratic cost at these levels is $(1\pm\varepsilon)^2\beta_B$. The Gaussian norm bound used in (4.11) gives the upper rate. For the lower rate, take a minimizer $\boldsymbol{x}_*$ in (2.12) and a small ball D about $(1+\varepsilon)\boldsymbol{x}_* + \delta\mathbf{1}$, with radius less than δ. This ball is feasible for all sufficiently large z and

$$\mathbb{P}\{\boldsymbol{X}/z \in D\} = \frac{z^N}{(2\pi)^{N/2}\sqrt{\det\Sigma}} \int_D \exp\left\{-\frac{z^2}{2}(\boldsymbol{x}-\boldsymbol{\mu}/z)^\mathsf{T}\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}/z)\right\} d\boldsymbol{x}.$$

Bounding the integrand below by its infimum on D and then letting its radius, δ and ε tend to zero gives the lower rate; the factor z^N and the shift $\boldsymbol{\mu}/z$ do not affect it. Applying this claim at the marginal log-quantiles and at those quantiles minus $\log|A_\ell|$ proves (2.11).

For any feasible $\boldsymbol{x}$ in (2.12), select $a \in A_\ell$ and $b \in A_r$ meeting the constraints of two distinct groups $\ell, r \in B$. Cauchy–Schwarz with $\boldsymbol{w} = \boldsymbol{e}_a/\sigma_a + \boldsymbol{e}_b/\sigma_b$ gives

$$\boldsymbol{x}^\mathsf{T}\Sigma^{-1}\boldsymbol{x} \ge \frac{(x_a/\sigma_a + x_b/\sigma_b)^2}{2(1+\rho_{ab})} \ge \frac{2}{1+\rho_{ab}} > 1.$$

There are finitely many pairs, so $\beta_B > 1$ for $|B| \ge 2$. For $B = \{\ell\}$, the analogous bound is one and is attained by $\boldsymbol{x} = \Sigma_{I_N,\{a\}}/\sigma_a$ with $a \in A_\ell$ and $\sigma_a = \sigma_\ell^*$. If the variances are constant within two groups, the displayed pairwise bound is attained by the Gaussian regression vector with selected coordinates (σ_a, σ_b). Minimizing over the pairs proves (2.13).

For $\boldsymbol{x} \in [0,\infty)^d$, at marginal levels qx_ℓ, every pairwise intersection is bounded by $J_{\{\ell,r\},\boldsymbol{T}}(q\max_{1\le i\le d} x_i) = o(q)$ when $\max_{1\le i\le d} x_i > 0$. Bonferroni's inequalities therefore yield

$$\lim_{q\downarrow 0} \frac{1 - C_{\boldsymbol{T}}(\mathbf{1} - q\boldsymbol{x})}{q} = \sum_{\ell=1}^d x_\ell, \qquad \boldsymbol{x} \in [0,\infty)^d. \tag{4.18}$$

□

Proof of Corollary 2.4. Fix $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$. For $i \in \{1,2\}$, the scalar case of Theorem 2.1 gives

$$1 - F_{i,\boldsymbol{T}}(u) \sim n_i \overline{\Phi}\left(\frac{\log u - m_i}{\sigma_i^*}\right), \qquad u \to \infty. \tag{4.19}$$

Now let $q \downarrow 0$ and put $z = \overline{\Phi}^{-1}(q)$, so $z \to \infty$. Mills' ratio and inversion yield

$$y_i := \frac{\log F_{i,\boldsymbol{T}}^{-1}(1-q) - m_i}{\sigma_i^*} = z + \frac{\log n_i}{z} + o(z^{-1}). \tag{4.20}$$

Standardize the coordinates as $Z_j = (X_j - m_i)/\sigma_i^*$, $i \in \{1,2\}$, $j \in A_i$. For $(a,b) \in A_1 \times A_2$ with correlation ρ_*, the minimum quadratic cost at thresholds one is $1/(1+\rho_*)$ and every other coordinate in these two groups has minimizing value

$$\frac{\rho_{ja} + \rho_{jb}}{1+\rho_*} < 1, \qquad j \in (A_1 \cup A_2) \setminus \{a,b\}.$$

Here the within-group correlation is strictly less than one and the cross-group correlation is at most ρ_*. Pairs with smaller correlation already have a larger cost. Hence every triple meeting both groups has a strictly larger cost than $1/(1+\rho_*)$. By finiteness, for some $\delta, \varepsilon > 0$, the probability that such a triple exceeds $(1-\delta)z$ is $O(e^{-[1/(1+\rho_*)+\varepsilon]z^2})$.
Outside this event, a joint aggregate exceedance requires a unique coordinate above $(1-\delta)z$ in each group, since $T_i \le n_i \exp(m_i + \sigma_i^* \max_{j\in A_i} Z_j)$. The other summands perturb the standardized thresholds by $O(e^{-\sigma_i^* \delta z}) = o(z^{-1})$. The orthant equivalent (4.5) is stable under these perturbations; distinct pair intersections require a triple exceedance. Thus the joint tail is asymptotic to the sum of the pair tails with correlation ρ_*. By (4.5) and (4.20), each leading pair contributes

$$\frac{(1+\rho_*)^2}{2\pi z^2 \sqrt{1-\rho_*^2}} e^{-z^2/(1+\rho_*)} (n_1 n_2)^{-1/(1+\rho_*)} (1 + o(1)).$$

Summing the n_* contributions and using $q \sim e^{-z^2/2}/(\sqrt{2\pi} z)$ and $z^2 \sim 2\log(1/q)$ proves (2.14)–(2.15). □

Proof of Corollary 2.5. Fix $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$ and write σ^2 for the common variance, $\alpha_0 = 2/(1+\rho_*) > 1$ and $\widehat{C}_{\boldsymbol{T}}(u,v) = u + v - 1 + C_{\boldsymbol{T}}(1-u, 1-v)$. For $i \in \{1,2\}$, put $\mu_i^* = \max_{j\in A_i} \mu_j$ and $k_i = \#\{j \in A_i : \mu_j = \mu_i^*\}$. The scalar equivalent is $1 - F_{i,\boldsymbol{T}}(u) \sim k_i \overline{\Phi}((\log u - \mu_i^*)/\sigma)$ as $u \to \infty$. Now let $q \downarrow 0$ and put $z = \overline{\Phi}^{-1}(q)$, so $z \to \infty$. Mills' ratio and inversion therefore imply for each fixed $x > 0$

$$\log F_{i,\boldsymbol{T}}^{-1}(1-qx) = \mu_i^* + \sigma\left(z + \frac{\log k_i - \log x}{z} + o(z^{-1})\right), \qquad i \in \{1,2\}.$$

Consequently, for $i \in \{1,2\}$ and $j \in A_i$,

$$\log F_{i,\boldsymbol{T}}^{-1}(1-q) - \mu_j = \sigma z + O(1), \qquad \log F_{i,\boldsymbol{T}}^{-1}(1-qx) - \log F_{i,\boldsymbol{T}}^{-1}(1-q) = -\sigma \frac{\log x}{z} + o(z^{-1}).$$

Put $Z_j = (X_j - \mu_j)/\sigma$. The bound $T_i \le |A_i| \exp(\mu_i^* + \sigma \max_{j\in A_i} Z_j)$ gives the same pair localization as above: bounded threshold shifts change the Gaussian exponents by $O(z)$, whereas the gaps for smaller correlations and triple exceedances have order z^2. Thus the joint tail is asymptotic to the sum over pairs (a,b) with $\rho_{ab} = \rho_*$. Among these pairs, those maximizing $\mu_a + \mu_b$ dominate at the linear order in z. Nevertheless, (4.5) shows that replacing q, q by qx, qy multiplies every such pair probability by $(xy)^{1/(1+\rho_*)}(1+o(1))$: the standardized threshold changes are $-(\log x)/z + o(z^{-1})$ and $-(\log y)/z + o(z^{-1})$, while each component of the standardized pair's Gaussian precision vector is $z/(1+\rho_*) + O(1)$. The common multiplier therefore survives the finite sum, irrespective of the pairs' relative weights. Consequently, we have

$$\frac{\widehat{C}_{\boldsymbol{T}}(qx, qy)}{\widehat{C}_{\boldsymbol{T}}(q,q)} \longrightarrow (xy)^{\alpha_0/2}, \qquad q \downarrow 0, \quad x, y > 0.$$

Taking $x = y = c$ shows that the diagonal tail is regularly varying at zero with index α_0. Choose b_0 so that $\lim_{r\to\infty} r\widehat{C}_{\boldsymbol{T}}(1/b_0(r), 1/b_0(r)) = 1$. Then b_0 is regularly varying with index $1/\alpha_0$ and $\lim_{r\to\infty} b_0(r)/r = 0$, while the ratio limit gives

$$\lim_{r\to\infty} r\mathbb{P}\{W_{1,\boldsymbol{T}} > b_0(r)x,\ W_{2,\boldsymbol{T}} > b_0(r)y\} = \nu_0((x,\infty)\times(y,\infty)) = (xy)^{-\alpha_0/2}, \qquad x, y > 0.$$

For each fixed $\varepsilon > 0$ and $M > 0$, the limsup as $r \to \infty$ of the normalized mass where $\min(w_1, w_2) > \varepsilon$ and $\max(w_1, w_2) > M$ is at most $2(M\varepsilon)^{-\alpha_0/2}$, which tends to zero as $M \to \infty$. On each compact set bounded away from the axes, the rectangle limits are convergence-determining. The preceding bound controls the mass escaping to infinity and proves M_0-convergence on $(0,\infty)^2$, that is, convergence against bounded continuous functions vanishing on a neighbourhood of the axes. The limit is homogeneous of order $-\alpha_0$. Finally, standard Pareto margins and (4.18) give ordinary regular variation with index one and mass on the axes. □

Proof of Remark 2.6. Let $q \downarrow 0$ and put $u = F^{-1}_{1,\boldsymbol{Q}}(1-q)$ and $t_u = \log u$, so $\lim_{q\downarrow 0} u = \lim_{q\downarrow 0} t_u = \infty$. For suitable $\sigma > 0$, $M \in \mathbb{R}$ and a positive integer k, the Gaussian maximum in the first group satisfies

$$\mathbb{P}\{Q_1 > e^v\} \sim k\overline{\Phi}((v-M)/\sigma) = e^{\psi(v)}, \qquad \psi'(v) \sim -v/\sigma^2, \qquad v \to \infty.$$

By the hypothesis and the scalar case of Theorem 2.1, this equivalent holds for every T_ℓ, $\boldsymbol{T} \in \{\boldsymbol{S}, \boldsymbol{Q}\}$. Hence, writing $u_{\ell,\boldsymbol{T}} = F^{-1}_{\ell,\boldsymbol{T}}(1-q)$, we have $\lim_{q\downarrow 0}[\psi(\log u_{\ell,\boldsymbol{T}}) - \psi(t_u)] = 0$. Mills' ratio and the mean value theorem give $\log u_{\ell,\boldsymbol{T}} = t_u + o(t_u^{-1})$ for all $2d$ quantiles. Let u_- and u_+ be their minimum and maximum. Then

$$\mathbb{P}\{\boldsymbol{T} > u_+\mathbf{1}\} \le J_{B,\boldsymbol{T}}(q) \le \mathbb{P}\{\boldsymbol{T} > u_-\mathbf{1}\}, \qquad B = \{1,\dots,d\}.$$

Since $\log u_\pm = t_u + o(t_u^{-1})$, (2.7) and the stability (4.9) show that both bounds are asymptotic to $\mathbb{P}\{\boldsymbol{T} > u\mathbf{1}\}$. Taking the ratio for sums and maxima and using (1.3) proves (2.16). □

Proof of Corollary 2.7. Fix a group ℓ and let $p \downarrow 0$. Put

$$A_p = \{Q_\ell > F^{-1}_{\ell,\boldsymbol{Q}}(1-p)\}, \quad B_p = \{S_\ell > F^{-1}_{\ell,\boldsymbol{S}}(1-p)\}, \quad D_p = \{S_\ell > F^{-1}_{\ell,\boldsymbol{Q}}(1-p)\}.$$

Since $Q_\ell \le S_\ell$, their quantiles are ordered and $A_p, B_p \subset D_p$. Continuity gives $\mathbb{P}\{A_p\} = \mathbb{P}\{B_p\} = p$, while the scalar case of Theorem 2.1 implies $\mathbb{P}\{D_p\} = p + o(p)$ and hence

$$\mathbb{P}\{A_p \triangle B_p\} \le 2(\mathbb{P}\{D_p\} - p) = o(p).$$

For fixed $\boldsymbol{x}, \boldsymbol{y}$, we can therefore replace each sum exceedance at marginal probability level qx_ℓ by the corresponding maximum exceedance, with total error $o(q)$; zero coordinates are omitted. With $z_\ell = \max(x_\ell, y_\ell)$ this yields

$$C_{\boldsymbol{S},\boldsymbol{Q}}(\mathbf{1} - q\boldsymbol{x}, \mathbf{1} - q\boldsymbol{y}) = C_{\boldsymbol{Q}}(\mathbf{1} - q\boldsymbol{z}) + o(q).$$

Equation (4.18) proves (2.17). Its first-order expansion also gives

$$\lim_{n\to\infty} C_{\boldsymbol{S},\boldsymbol{Q}}(\boldsymbol{u}^{1/n}, \boldsymbol{v}^{1/n})^n = e^{-\sum_{\ell=1}^d \max(-\log u_\ell, -\log v_\ell)} = \prod_{\ell=1}^d \min(u_\ell, v_\ell)$$

for $\boldsymbol{u}, \boldsymbol{v} \in (0,1]^d$, proving (2.18). □

Proof of Examples 3.1 and 3.2 and Remark 3.3. In Example 3.1, the only admissible set is $I = \{1\}$, with $\boldsymbol{s}^I = \mathbf{1}_3$ and $\kappa = 1/2$. Indeed, $s^I_1 = 1/2$ for either other singleton, whereas a pair containing coordinate 1 has a zero multiplier. The boundary variables have law (Y_1, Y_2), so their maximum exceeds zero with probability $1 - \Phi(0)^2 = 3/4$. Formula (2.8) gives the stated constant. Its strict bounds follow because $\{\max(Y_1, Y_2) > 0\}$ is a proper subset up to positive probability of $\{e^{Y_1} + e^{Y_2} > 1\}$, whose complement also has positive probability.

For distinct coordinate–time pairs in Example 3.2

$$\mathrm{Cov}(X_i(k), X_j(l)) = \tfrac{1}{4} + \tfrac{1}{2}\cos((k-l)\pi/3).$$

The independent noise contributes one quarter of the identity matrix to each finite sampled covariance matrix, proving positive definiteness. Singletons are infeasible; pairs I at adjacent times have $s^I_j = 1/3$ for

both coordinates j at the remaining time and are also infeasible. Each one-per-time triple has multiplier vector $(1,0,1)^\mathsf{T}$. Thus $\mathfrak{I}_*$ consists of the four pairs selecting one coordinate at time 0 and one at time 2. Each has $\kappa = 1$ and its only boundary group is at time 1. Its two conditional variables, denoted by $\widetilde{Y}_1, \widetilde{Y}_2$, have mean zero and covariance

$$\begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{pmatrix}.$$

Consequently, $\mathbb{P}\{\max(\widetilde{Y}_1, \widetilde{Y}_2) > 0\} = 1 - (1/4 + \arcsin(1/2)/(2\pi)) = 2/3$ and (2.8) yields $C_* = \frac{3}{2}\mathbb{P}\{e^{\widetilde{Y}_1} + e^{\widetilde{Y}_2} > 1\} \in (1, 3/2)$. For any two time groups, Remark 2.2 and Corollary 2.4 give the remaining claims. In Remark 3.3, every admissible set is a cross-group pair with covariance c, multiplier vector $(v+c)^{-1}\mathbf{1}_2$ and cost $(v+c)^{-1}$. Hence exactly the n_* pairs with covariance c_* are dominant. They have identical Gaussian laws and no boundary groups, so Theorem 2.1 gives (3.5); Corollary 2.4 gives the asserted copula equivalent. □

Proof of Example 3.4. For a selector I with $|I| = k$, the equicorrelated matrix gives

$$\boldsymbol{\lambda}_I = \frac{\mathbf{1}_I}{\sigma^2 D_k}, \qquad s_j^I = \frac{k\rho}{D_k} < 1 \quad (j \notin I),$$

since $D_k > 0$ and $D_k - k\rho = 1 - \rho > 0$. Thus an unrepresented group cannot satisfy (2.2). The admissible sets are exactly the selectors containing one coordinate from each group and they all have cost $d/(2\sigma^2 D_d)$ and no boundary groups. Their selected Gaussian vectors have identical laws. Theorem 2.1 and (4.5), using

$$\det\left[(1-\rho)\mathrm{Id}_d + \rho \mathbf{1}\mathbf{1}^\mathsf{T}\right] = (1-\rho)^{d-1} D_d,$$

prove $C_* = 1$ and (3.6). Applying the same calculation to the groups in B gives the minimum standardized quadratic cost b/D_b in (2.12), proving (3.7). □

Proof of Proposition 3.5. Let $u \to \infty, t_u = \log u$ with $\lim_{u\to\infty} t_u = \infty$. Write $q(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^\mathsf{T}\Sigma^{-1}\boldsymbol{x}$ and $\kappa_\pm = \min_{\mathcal{F}_\pm} q$. The set $\mathcal{F}_\pm$ is the union over $p_\ell \in P_\ell$ of the polyhedra

$$\{\boldsymbol{x} : x_{p_\ell} \geq 1 \ (1 \leq \ell \leq d), \quad x_{p_\ell} \geq x_k \ (1 \leq \ell \leq d, \ k \in N_\ell)\}.$$

Each has a unique quadratic minimizer, so $\mathcal{S}_\pm$ is finite and nonempty. Fix $\boldsymbol{s} \in \mathcal{S}_\pm$ and select $s_{p_\ell} = h_\ell$. The optimality conditions give

$$\boldsymbol{g} = \sum_{1\leq \ell \leq d} \left[\beta_\ell \boldsymbol{e}_{p_\ell} + \sum_{k \in N_\ell} \nu_{\ell k}(\boldsymbol{e}_{p_\ell} - \boldsymbol{e}_k) \right], \qquad \beta_\ell, \nu_{\ell k} \geq 0, \tag{4.21}$$

with $\beta_\ell = \sum_{j \in A_\ell} g_j$ and complementary slackness. Put $B = \{\ell : \boldsymbol{g}_{A_\ell} \neq \mathbf{0}\}$. For $\ell \in B$, $g_{p_\ell} > 0$ and p_ℓ is the unique positive maximizer: choosing another maximizer in its place would force $g_{p_\ell} = 0$ in (4.21). Define

$$K_\ell = \{k \in N_\ell : \nu_{\ell k} > 0\}, \qquad a_\ell = g_{p_\ell} = \beta_\ell + \sum_{k \in K_\ell} \nu_{\ell k}.$$

Complementary slackness gives $s_k = h_\ell$ for $k \in K_\ell$ and $\beta_\ell(h_\ell - 1) = 0$. For $j \in P_\ell \setminus \{p_\ell\}$, uniqueness of the positive maximizer gives $s_j < h_\ell$. For $k \in N_\ell \setminus K_\ell$, (4.21) gives $g_k = -\nu_{\ell k} = 0$, so the second implication in (3.13) gives $s_k < h_\ell$. Thus for $\ell \in B$ we have $s_j < h_\ell$ whenever $j \in A_\ell \setminus (\{p_\ell\} \cup K_\ell)$. With

$$B^1 = \{\ell \in B : h_\ell = 1\}, \qquad B^> = \{\ell \in B : h_\ell > 1\},$$

the first implication in (3.13) gives $\beta_\ell > 0$ on B^1, whereas complementary slackness gives $a_\ell = \sum_{k \in K_\ell} \nu_{\ell k} > 0$ on $B^>$. Moreover, $2\kappa_\pm = \boldsymbol{g}^\mathsf{T}\boldsymbol{s} = \sum_{\ell \in B} \beta_\ell > 0$, so $B^1 \neq \varnothing$.
Let $L\boldsymbol{X}$ collect $(X_{p_\ell}, (X_k)_{k\in K_\ell})$ for $\ell \in B^1$ and $(X_k - X_{p_\ell})_{k \in K_\ell}$ for $\ell \in B^>$. Its n rows are linearly independent. Set next

$$K = L\Sigma L^\mathsf{T}, \qquad \boldsymbol{m} = L\boldsymbol{\mu}, \qquad \boldsymbol{c} = L\boldsymbol{s}, \qquad \boldsymbol{v} = K^{-1}\boldsymbol{c}.$$

In this notation, the corresponding subvectors are

$$(\boldsymbol{c}_\ell, \boldsymbol{v}_\ell) = \begin{cases} ((1,\mathbf{1}), (a_\ell, (-\nu_{\ell k})_{k\in K_\ell})), & \ell \in B^1, \\ (\mathbf{0}, (-\nu_{\ell k})_{k\in K_\ell}), & \ell \in B^>. \end{cases}$$

Indeed, (4.21) gives $\boldsymbol{g} = L^\mathsf{T}\boldsymbol{v}$ and hence

$$\boldsymbol{s} = \Sigma L^\mathsf{T} K^{-1}\boldsymbol{c}, \qquad \tfrac{1}{2}\boldsymbol{c}^\mathsf{T} K^{-1}\boldsymbol{c} = \kappa_\pm. \tag{4.22}$$

Let $H_s(u)$ be the event

$$\begin{cases} e^{X_{p_\ell}} - \sum_{k\in K_\ell} e^{X_k} > u, & \ell \in B^1, \\ \sum_{k\in K_\ell} e^{X_k - X_{p_\ell}} < 1, & \ell \in B^>. \end{cases}$$

Within each group factor below, write $p = p_\ell$ and let every index k range over K_ℓ; y_{k-p} denotes the entry corresponding to $X_k - X_{p_\ell}$. Writing $L\boldsymbol{X} = t_u\boldsymbol{c} + \boldsymbol{y}$ gives a fixed product domain $\mathcal{D}$: its factors are $e^{y_p} - \sum_k e^{y_k} > 1$ on B^1 and $\sum_k e^{y_{k-p}} < 1$ on $B^>$. Therefore we obtain

$$\mathbb{P}\{H_s(u)\} = \frac{e^{-\kappa_\pm t_u^2 + t_u \boldsymbol{v}^\mathsf{T}\boldsymbol{m}}}{(2\pi)^{n/2}\sqrt{\det K}} \int_{\mathcal{D}} e^{-t_u \boldsymbol{v}^\mathsf{T}\boldsymbol{y}} e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{m})^\mathsf{T} K^{-1}(\boldsymbol{y}-\boldsymbol{m})} d\boldsymbol{y}. \tag{4.23}$$

The first factor integrates to

$$\int_{\mathcal{D}} e^{-t_u \boldsymbol{v}^\mathsf{T}\boldsymbol{y}} d\boldsymbol{y} = \prod_{\ell\in B^1} \frac{\Gamma(t_u\beta_\ell)\prod_{k\in K_\ell}\Gamma(t_u\nu_{\ell k})}{\Gamma(t_u a_\ell + 1)} \prod_{\ell\in B^>} \frac{\prod_{k\in K_\ell}\Gamma(t_u\nu_{\ell k})}{\Gamma(t_u a_\ell + 1)}, \tag{4.24}$$

which follows by substituting $z_k = e^{y_k}$ on B^1 and integrating y_p first. Under the resulting normalized measure, $z_k = G_k/G_0$, with independent unit-scale gamma variables of shapes $t_u\nu_{\ell k}$ and $t_u\beta_\ell$, while $y_p = \log(1 + \sum_k z_k) + E/(t_u a_\ell)$ with E standard exponential. On $B^>$, $z_k = e^{y_{k-p}}$ and their slack have Dirichlet parameters $((t_u\nu_{\ell k})_k, 1)$. Thus the normalized measure in (4.24) concentrates at $\boldsymbol{b}$, whose group entries are

$$\boldsymbol{b}_\ell = \begin{cases} (\log(a_\ell/\beta_\ell), (\log(\nu_{\ell k}/\beta_\ell))_k), & \ell \in B^1, \\ (\log(\nu_{\ell k}/a_\ell))_k, & \ell \in B^>. \end{cases}$$

The Gaussian factor in (4.23) is bounded, continuous and positive at $\boldsymbol{b}$. Stirling's formula in (4.24) therefore yields as $u \to \infty$

$$\mathbb{P}\{H_s(u)\} \sim A_s t_u^{-m_s} e^{-\kappa_\pm t_u^2 - \alpha_s t_u}, \qquad m_s = (n + |B|)/2, \quad \alpha_s = \boldsymbol{v}^\mathsf{T}(\boldsymbol{b} - \boldsymbol{m}), \quad A_s > 0. \tag{4.25}$$

More explicitly we have

$$A_s = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{b}-\boldsymbol{m})^\mathsf{T} K^{-1}(\boldsymbol{b}-\boldsymbol{m})\right)}{(2\pi)^{|B|/2}\sqrt{\det K}} \prod_{\ell\in B} \left(a_\ell \prod_{k\in K_\ell} \nu_{\ell k}\right)^{-1/2} \prod_{\ell\in B^1} \beta_\ell^{-1/2},$$

where empty products equal one. The same concentration argument gives $L\boldsymbol{X} - t_u\boldsymbol{c} \xrightarrow[u\to\infty]{\mathbb{P}} \boldsymbol{b}$ conditionally on $H_s(u)$. Gaussian regression and (4.22) now imply

$$\mathrm{Law}(\boldsymbol{X} - t_u\boldsymbol{s} \mid H_s(u)) \xrightarrow[u\to\infty]{\mathrm{w}} \mathrm{Law}(\boldsymbol{W}_s), \qquad \mathrm{Law}(\boldsymbol{W}_s) = \mathrm{Law}(\boldsymbol{X} \mid L\boldsymbol{X} = \boldsymbol{b}). \tag{4.26}$$

In particular, $\boldsymbol{X}/t_u \xrightarrow[u\to\infty]{\mathbb{P}} \boldsymbol{s}$ under this conditioning.
Choose small disjoint neighbourhoods U_s of the points in $\mathcal{S}_\pm$. A signed exceedance implies

$$\max_{j\in P_\ell} X_j \geq t_u - \log|P_\ell|, \qquad \max_{j\in P_\ell} X_j \geq \max_{k\in N_\ell} X_k - \log|P_\ell|.$$

Coercivity gives a strict quadratic-cost gap outside $\bigcup_s U_s$. The Gaussian norm bound used in (4.11), together with (4.25), makes this contribution $o(\mathbb{P}\{H_s(u)\})$ for every s.

Fix s and put $V_s(u)=\{\boldsymbol{X}/t_u\in U_s\}$. For $\ell\in B$, write

$$O_\ell=A_\ell\setminus(\{p_\ell\}\cup K_\ell),\quad R_\ell=\sum_{j\in O_\ell}e^{X_j},$$
$$M_\ell=e^{X_{p_\ell}}-\sum_{k\in K_\ell}e^{X_k},\quad Z_\ell=\sum_{k\in K_\ell}e^{X_k-X_{p_\ell}}.$$

The inequalities $s_j<h_\ell$ for $j\in O_\ell$, $\ell\in B$ and $h_\ell>1$ for $\ell\in B^>$ allow us to shrink U_s and choose $\delta>0$ so that for every $\boldsymbol{x}\in U_s$ the coordinate x_{p_ℓ} is the largest among $(x_j)_{j\in P_\ell}$ in each active group and the following inequalities hold:

$$\begin{cases} x_j\le 1-\delta, & \ell\in B^1,\ j\in O_\ell,\\ x_{p_\ell}\ge 1+\delta, & \ell\in B^>,\\ x_j-x_{p_\ell}\le-\delta, & \ell\in B^>,\ j\in O_\ell.\end{cases}$$

Consequently, on $V_s(u)$, the following bounds hold deterministically for either $\widetilde{\boldsymbol{T}}\in\{\widetilde{\boldsymbol{S}},\widetilde{\boldsymbol{Q}}\}$:

$$|\widetilde{T}_\ell-M_\ell|\le R_\ell\quad(\ell\in B),$$
$$\begin{cases} R_\ell/u\le Ne^{-\delta t_u}, & \ell\in B^1,\\ (R_\ell+u)e^{-X_{p_\ell}}\le(N+1)e^{-\delta t_u}, & \ell\in B^>.\end{cases}$$

Set $\varepsilon_u=(N+1)e^{-\delta t_u}<1$ for large u and define

$$H_s^-(u)=\{M_\ell>u(1+\varepsilon_u),\ \ell\in B^1;\ Z_\ell<1-\varepsilon_u,\ \ell\in B^>\},$$
$$H_s^+(u)=\{M_\ell>u(1-\varepsilon_u),\ \ell\in B^1;\ Z_\ell<1+\varepsilon_u,\ \ell\in B^>\},\quad E_{B,\widetilde{\boldsymbol{T}}}(u)\ =\{\widetilde{T}_\ell>u,\ \ell\in B\}.$$

The deterministic bounds give the event inclusions

$$V_s(u)\cap H_s^-(u)\subset V_s(u)\cap E_{B,\widetilde{\boldsymbol{T}}}(u)\subset V_s(u)\cap H_s^+(u),$$
$$H_s^-(u)\subset H_s(u)\subset H_s^+(u).$$

In the $\boldsymbol{y}$ coordinates of (4.23), the domains for $H_s^\pm(u)$ are $\mathcal{D}+\boldsymbol{\tau}_u^\pm$. On a B^1 block, all entries of $\boldsymbol{\tau}_u^-$ and $\boldsymbol{\tau}_u^+$ are $\log(1+\varepsilon_u)$ and $\log(1-\varepsilon_u)$; on a $B^>$ block, these shifts are interchanged. Thus $\|\boldsymbol{\tau}_u^\pm\|=O(\varepsilon_u)$ and $\lim_{u\to\infty}t_u\varepsilon_u=0$. Translating back to $\mathcal{D}$ in (4.23), we have $\lim_{u\to\infty}e^{-t_u\boldsymbol{v}^{\mathsf{T}}\boldsymbol{\tau}_u^\pm}=1$; the bounded Gaussian factor has the same limit at $\boldsymbol{b}$ and hence

$$\lim_{u\to\infty}\frac{\mathbb{P}\{H_s^\pm(u)\}}{\mathbb{P}\{H_s(u)\}}=1.$$

Together with (4.26), the inclusions imply as $u\to\infty$

$$\mathbb{P}\Big\{(V_s(u)\cap E_{B,\widetilde{\boldsymbol{T}}}(u))\triangle H_s(u)\Big\}\le\mathbb{P}\Big\{H_s^+(u)\setminus H_s^-(u)\Big\}+\mathbb{P}\{H_s(u)\cap V_s(u)^c\}=o(\mathbb{P}\{H_s(u)\}).$$

This estimate remains valid after intersecting both events with the exceedance conditions in the inactive groups.

For $\ell\in\{1,\dots,d\}\setminus B$, put $J_\ell^+=\{j\in P_\ell: s_j=h_\ell\}$ and $J_\ell^-=\{k\in N_\ell: s_k=h_\ell\}$. The boundary probabilities obtained from (4.26) are

$$C_s=\mathbb{P}\left\{\sum_{j\in J_\ell^+}e^{W_{s,j}}-\sum_{k\in J_\ell^-}e^{W_{s,k}}>\mathbb{I}_{\{h_\ell=1\}},\quad \ell\in\{1,\dots,d\}\setminus B\right\},$$
$$D_s=\mathbb{P}\left\{\max_{j\in J_\ell^+}e^{W_{s,j}}-\sum_{k\in J_\ell^-}e^{W_{s,k}}>\mathbb{I}_{\{h_\ell=1\}},\quad \ell\in\{1,\dots,d\}\setminus B\right\}.$$

If $B=\{1,\dots,d\}$, then we set $C_s=D_s=1$. The coordinates of $\boldsymbol{W}_s$ in these groups have positive definite covariance, because L only involves active groups. Their exceedance boundaries are null: condition on all but one positive coordinate for sums, or all but one negative coordinate for adjusted maxima;

without negative terms, use continuity of the Gaussian maximum. Full support and event inclusion give $0 < D_s \le C_s \le 1$. Thus localization and (4.26) yield as $u \to \infty$

$$\mathbb{P}\Big\{\widetilde{\boldsymbol{S}} > u\mathbf{1}\Big\} \sim \sum_{s\in\mathcal{S}_\pm} C_s \mathbb{P}\{H_s(u)\}, \qquad \mathbb{P}\Big\{\widetilde{\boldsymbol{Q}} > u\mathbf{1}\Big\} \sim \sum_{s\in\mathcal{S}_\pm} D_s \mathbb{P}\{H_s(u)\}.$$

Let $\mathfrak{H}_\pm$ consist of the minimizers in $\mathcal{S}_\pm$ with smallest α_s and, among these, smallest m_s. By (4.25), only these terms contribute to the two finite sums implying

$$C_\pm = \frac{\sum_{s\in\mathfrak{H}_\pm} A_s C_s}{\sum_{s\in\mathfrak{H}_\pm} A_s D_s} \in [1,\infty).$$

For $d = 1$, $B = \{1\}$, so $C_s = D_s = 1$ for every minimizer and $C_\pm = 1$. Without negative terms, $B^> = \varnothing$, $K_\ell = \varnothing$, $a_\ell = \beta_\ell$ and $\boldsymbol{b} = \mathbf{0}$. Then L selects $I = \{p_\ell : \ell \in B\}$ and the above quantities become $A_s = A_I$, $\alpha_s = -\boldsymbol{\lambda}_I^\top \boldsymbol{\mu}_I$ and $m_s = |I|$. The correspondence $\boldsymbol{s} = \boldsymbol{s}^I$ identifies $\mathfrak{H}_\pm$ with $\mathfrak{H}$ and the displayed ratio reduces to (2.8).

□

Proof of (3.8), (3.9) *and* (3.10). Let $u \to \infty$, put $t_u = \log u$ and set $f(u) = \mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\}$. By (4.16) and (4.17), for constants $K, m > 0$ and $\gamma \in \mathbb{R}$

$$f(e^s) \sim \Psi(s) := K s^{-m} e^{-\kappa s^2 + \gamma s}, \qquad s \to \infty.$$

Since $\{S_\wedge > v\} = \{\boldsymbol{S} > v\mathbf{1}\}$, for each fixed $x \ge 0$

$$\mathbb{P}\Big\{\frac{2\kappa t_u}{u}(S_\wedge - u) > x \mid \boldsymbol{S} > u\mathbf{1}\Big\} = \frac{f(u(1 + x/(2\kappa t_u)))}{f(u)} \xrightarrow[u\to\infty]{} e^{-x}.$$

This proves the first claim in (3.10) and gives $S_\wedge/u \xrightarrow[u\to\infty]{\mathbb{P}} 1$ conditionally on $\boldsymbol{S} > u\mathbf{1}$. Applying Tonelli's theorem we also obtain

$$\mathbb{E}\{S_\wedge - u \mid \boldsymbol{S} > u\mathbf{1}\} = \int_{t_u}^\infty e^s f(e^s)\, ds / f(u).$$

The tail equivalent gives the uniform relation

$$\lim_{u\to\infty} \sup_{s \ge t_u} \left| \frac{f(e^s)}{\Psi(s)} - 1 \right| = 0.$$

Substitution $s = t_u + z/t_u$ yields

$$\frac{t_u \int_{t_u}^\infty e^s \Psi(s)\, ds}{u\Psi(t_u)} = \int_0^\infty (1 + z/t_u^2)^{-m} e^{-2\kappa z - \kappa z^2/t_u^2 + (\gamma+1)z/t_u}\, dz \xrightarrow[u\to\infty]{} \frac{1}{2\kappa},$$

since the integrand is bounded by $e^{-\kappa z}$ for large u, hence the second claim in (3.10) follows. For signed sums (4.25) and the finite mixtures in the proof of Proposition 3.5 yield the same form of Ψ, with $\kappa_\pm$ replacing κ, $m = m_s > 0$ and $\gamma = -\alpha_s$ on the dominant set. The same argument proves the signed extension.

For (3.8) write $Q_\wedge = \min_\ell Q_\ell$ and $E_u = \{\boldsymbol{S} > u\mathbf{1}\}$. In view of (4.13) and (2.6) for each fixed $r > 1$ we obtain as $u \to \infty$

$$\mathbb{P}\{E_u, Q_\wedge > u/r\} \sim \sum_{I\in\mathfrak{I}_*} p_I(u) \mathbb{P}\{\boldsymbol{S}^{(I)} > \mathbf{1}, \boldsymbol{Q}^{(I)} > r^{-1}\mathbf{1}\}.$$

Indeed, represented groups exceed u apart from (4.12) and unrepresented groups with $\max_{j\in A_\ell} s_j^I > 1$ do so with conditional probability tending to one. Dividing by (4.17) gives the expression in (3.9) for the limiting conditional cdf of $u/Q_\wedge$ at $r > 1$. It is continuous there and tends to one as $r \to \infty$. For each fixed $0 < r < 1$ the corresponding probability is at most $\mathbb{P}\{\boldsymbol{Q} > (u/r)\mathbf{1}\}/f(u)$ and

$$\lim_{u\to\infty} \frac{\mathbb{P}\{\boldsymbol{Q} > (u/r)\mathbf{1}\}}{f(u)} = 0$$

by (4.16) and (4.17). Thus the limit is supported on $[1,\infty)$ and Slutsky's theorem, using $S_\wedge/u \xrightarrow[u\to\infty]{\mathbb{P}} 1$ under the conditioning, gives the same limit for R_u. Letting $r \downarrow 1$ in the limiting cdf, atomlessness of the boundary maxima and $\boldsymbol{Q}^{(I)} \le \boldsymbol{S}^{(I)}$ yield

$$\mathbb{P}\{\mathcal{R}=1\} = \frac{\sum_{I\in\mathfrak{H}} A_I \mathbb{P}\Big\{\boldsymbol{Q}^{(I)} > \mathbf{1}\Big\}}{\sum_{I\in\mathfrak{H}} A_I \mathbb{P}\{\boldsymbol{S}^{(I)} > \mathbf{1}\}} = C_*^{-1}$$

establishing the proof. □

Proof of (3.15), (3.16) *and* (3.17). Writing $f_{\boldsymbol{\mu}}$ for the Gaussian density, completion of the square gives

$$e^{x_a} f_{\boldsymbol{\mu}}(\boldsymbol{x}) = e^{\mu_a + \Sigma_{aa}/2} f_{\boldsymbol{\mu}+\Sigma \mathbf{e}_a}(\boldsymbol{x}).$$

Integrating over $\mathcal{E}_B(u)$, summing over $a \in A_\ell$ and conditioning proves (3.15).
For the remainder, use Remark 3.3, let $u \to \infty$ and put $t_u = \log u$, so $\lim_{u\to\infty} t_u = \infty$. Write $c_{kl} = \mathrm{Cov}(X_i(k), X_j(l))$. Applying the scalar case of Theorem 2.1 under each tilted mean yields as $u \to \infty$

$$\mathrm{MES}_i(u) \sim \frac{e^{\mu_i + v/2}}{n_j \overline{\Phi}((t_u-\mu_j)/\sqrt{v})} \sum_{k\in\mathcal{T}_i} \sum_{l\in\mathcal{T}_j} \overline{\Phi}\Big(\frac{t_u - \mu_j - c_{kl}}{\sqrt{v}}\Big).$$

Mills' ratio shows that the n_* pairs with $c_{kl} = c_*$ dominate and gives (3.16).
In order to prove (3.17) we consider next a leading Gaussian pair $\boldsymbol{Y} = (Y_1, Y_2)^\mathsf{T}$ with mean $(\mu_i, \mu_j)^\mathsf{T}$ and covariance $K = \left(\begin{smallmatrix} v & c_* \\ c_* & v \end{smallmatrix}\right)$. Put $\lambda = (v + c_*)^{-1} > 0$, $p(u) = \mathbb{P}\{\boldsymbol{Y} > t_u \mathbf{1}\}$ and $G(u) = \mathbb{E}\{(e^{Y_1} - u)_+ \mathbb{I}_{\{Y_2 > t_u\}}\}$. In the Gaussian density substitute $\boldsymbol{Y} = t_u \mathbf{1} + (x,y)^\mathsf{T}/t_u$, with $x, y > 0$. If f denotes this density and $\boldsymbol{g}(t_u) = K^{-1}(t_u \mathbf{1} - (\mu_i, \mu_j)^\mathsf{T})$, then

$$\frac{f(t_u \mathbf{1} + (x,y)^\mathsf{T}/t_u)}{f(t_u \mathbf{1})} = e^{-\frac{g_1(t_u)x + g_2(t_u)y}{t_u} - \frac{(x,y)K^{-1}(x,y)^\mathsf{T}}{2t_u^2}}.$$

Both components of $\boldsymbol{g}(t_u)/t_u$ converge to λ as $u \to \infty$. For fixed $x > 0$, $\lim_{u\to\infty} t_u(e^{x/t_u} - 1) = x$. Moreover, $t_u(e^{x/t_u} - 1) \le x e^{x/t_u}$, so for large u the integrand with this additional factor is bounded by $x e^{-\lambda x/4 - \lambda y/2}$. Dominated convergence in the numerator and denominator as $u \to \infty$ therefore yields

$$\lim_{u\to\infty} \frac{\log u}{u} \frac{G(u)}{p(u)} = \frac{\int_0^\infty \int_0^\infty x e^{-\lambda(x+y)}\, dx\, dy}{\int_0^\infty \int_0^\infty e^{-\lambda(x+y)}\, dx\, dy} = v + c_*. \tag{4.27}$$

Now apply the localization proof of Theorem 2.1 to the two groups. Their dominant configurations are the n_* pairs with covariance c_* and their common quadratic cost is $\kappa = (v + c_*)^{-1}$. In disjoint neighbourhoods of these configurations, each group has one selected coordinate and the omitted summands total at most $\varepsilon_u = C u e^{-\delta t_u}$, for fixed $C, \delta > 0$. The weighted contributions outside these neighbourhoods are negligible: bound the payoff by S_i and apply the density-tilting identity above to each summand. Each tilted Gaussian has the same covariance and hence the same strict quadratic-cost gap as in (4.10); the Gaussian norm bound gives $O(e^{-(\kappa+\eta)t_u^2})$, for some $\eta > 0$, which is $o(up(u)/t_u)$. The same argument controls the selected-pair payoff outside its own neighbourhood.
Within each neighbourhood, the aggregate payoff with the stress indicator is bounded below by the selected-pair payoff at u and above by that payoff at $u - \varepsilon_u$, with both pair thresholds shifted together. Since $\log(u - \varepsilon_u) = t_u + o(t_u^{-1})$ as $u \to \infty$, (4.9) and (4.27) yield $\lim_{u\to\infty} G(u - \varepsilon_u)/G(u) = 1$. Consequently, as $u \to \infty$

$$\mathbb{E}\Big\{(S_i - u)_+ \mathbb{I}_{\{S_j > u\}}\Big\} \sim n_* G(u) \sim (v + c_*) \frac{u}{t_u} \mathbb{P}\{\boldsymbol{S} > u\mathbf{1}\},$$

where the last step uses (3.5) establishing the proof. □

Proof of Proposition 3.6. Use (3.18) and retain the deterministic quantities in (4.1) and the regression coefficients in (2.4). Let $u \to \infty$ and put $t_u = \log u$. The free radial argument is denoted by ξ, while $r_I(t_u)$ is the radius determined by the log-threshold t_u. For $I \in \mathfrak{I}_*$, put $H_I(u) = \{\boldsymbol{X}_I > t_u \mathbf{1}_I\}$ and $v_0 = \sqrt{2\kappa}$

and define $h_I(t) = \sqrt{r_I(t)/w(r_I(t))}$ for sufficiently large real t. By (4.1) and the infinite-endpoint Gumbel condition

$$\lim_{u\to\infty} \frac{r_I(t_u)}{t_u} = v_0 > 0, \qquad \lim_{u\to\infty} r_I(t_u) = \infty, \quad \lim_{u\to\infty} r_I(t_u) w(r_I(t_u)) = \infty.$$

In view of [15, Theorems 3.1, 3.4 and Corollary 5.3] as $u \to \infty$

$$p_I(u) \sim \frac{\mathcal{B}_N(r_I(t_u))[r_I(t_u)/w(r_I(t_u))]^{|I|/2}}{(2\pi)^{|I|/2}\sqrt{\det \Sigma_{II}} \prod_{j\in I} g_{I,j}(t_u)}, \tag{4.28}$$

where

$$\mathcal{B}_N(\xi) = 2^{N/2-1}\Gamma(N/2)[\xi w(\xi)]^{1-N/2}\overline{F}(\xi),$$

and with $\boldsymbol{\Delta}_I, \boldsymbol{\eta}_I$ defined algebraically by (4.2)

$$\mathrm{Law}\left(w(r_I(t_u))\boldsymbol{\Delta}_I, \frac{\boldsymbol{\eta}_I}{h_I(t_u)} \mid H_I(u)\right) \xrightarrow[u\to\infty]{\mathrm{w}} \left(\bigotimes_{j\in I} \mathrm{Exp}(\lambda_{I,j}/v_0)\right) \otimes \mathcal{N}_{N-|I|}(\mathbf{0}, \Gamma_I). \tag{4.29}$$

In order to check the moving-threshold hypotheses we write next $\boldsymbol{z}_I(t_u) = t_u \mathbf{1}_I - \boldsymbol{\mu}_I$. The selected subvector has all its coordinates active for large u since

$$\Sigma_{II}^{-1}\boldsymbol{z}_I(t_u) = t_u \boldsymbol{\lambda}_I - \Sigma_{II}^{-1}\boldsymbol{\mu}_I > \mathbf{0}_I, \qquad \lim_{u\to\infty} \frac{\boldsymbol{z}_I(t_u)}{r_I(t_u)} = \frac{\mathbf{1}_I}{v_0}.$$

Thus the radius and scale in Theorem 3.1 of [15] are exactly $r_I(t_u)$ and $w(r_I(t_u))$; its limiting active multipliers are $\boldsymbol{\lambda}_I/v_0 > \mathbf{0}_I$. Applying the projection identity used in its Lemma 3.5 yields (4.28) in terms of the original N-dimensional radius R. For the conditional result, extend the centred threshold to I^c by $D_I\boldsymbol{z}_I(t_u)$. By [20, Lemma 1, Eqs. (20)–(21)], its minimal active set is I. The scaled residual threshold in condition (3.11) of [15] is zero, so its Corollary 5.3 applies. Replacing the deterministic regression centre $\boldsymbol{\mu}_{I^c} + D_I\boldsymbol{z}_I(t_u)$ by $\boldsymbol{\mu}_{I^c} + D_I(\boldsymbol{X}_I - \boldsymbol{\mu}_I)$ changes the normalized residual by $O_{\mathbb{P}}((r_I(t_u)w(r_I(t_u)))^{-1/2}) = o_{\mathbb{P}}(1)$ as $u \to \infty$, which proves (4.29) with (4.2). For $N = 1$, (4.28) reduces directly to $p_I(u) = \overline{F}(r_I(t_u))/2$ and the conditional limit is (3.19).

Both cases have $w \in \mathrm{RV}_a$ for some $a > -1$, with $a = 1$ in part (i). The Gumbel representation theorem [26, Chapter 1] implies for a sufficiently large fixed ξ_0

$$\overline{F}(\xi) = c(\xi)e - \int_{\xi_0}^{\xi} \widetilde{w}(s)\, ds, \qquad \xi \geq \xi_0, \tag{4.30}$$

where $\lim_{\xi\to\infty} c(\xi) = c_0 \in (0, \infty)$ and $\lim_{\xi\to\infty} \widetilde{w}(\xi)/w(\xi) = 1$. Consequently, by Karamata's theorem as $\xi \to \infty$

$$H(\xi) := -\log \overline{F}(\xi) \sim \int_{\xi_0}^{\xi} w(s)\, ds \sim \frac{\xi w(\xi)}{a+1}, \qquad H \in \mathrm{RV}_{a+1}\,. \tag{4.31}$$

Next we use the disjoint neighbourhoods and strict quadratic-cost gap from Lemma 4.1 and (4.10). Outside their union, a sum exceedance requires $R \geq (v_0 + \varepsilon)t_u - O(1)$ for some $\varepsilon > 0$. The last display and (4.28) make this contribution $o(p_I(u))$ as $u \to \infty$ for every $I \in \mathfrak{I}_*$. Moreover,

$$\lim_{u\to\infty} \frac{1}{t_u w(r_I(t_u))} = 0, \qquad \lim_{u\to\infty} \frac{h_I(t_u)}{t_u} = 0,$$

since these ratios equal $(r_I(t_u)/t_u)/(r_I(t_u)w(r_I(t_u)))$ and $(r_I(t_u)/t_u)/\sqrt{r_I(t_u)w(r_I(t_u))}$, respectively. Thus (4.29) gives $\boldsymbol{X}/t_u \xrightarrow[u\to\infty]{\mathbb{P}} \boldsymbol{s}^I$ conditionally on $H_I(u)$.

Inside a dominant neighbourhood, the unused terms in a represented group perturb its selected log-threshold by $O(e^{-\delta t_u})$. Since $\lim_{u\to\infty} w(r_I(t_u))e^{-\delta t_u} = 0$, the elliptical orthant equivalent is stable under this perturbation. Thus the local reduction (4.13) holds for sums and maxima under the elliptical law. Unrepresented groups with $\max_{j\in A_\ell} s_j^I > 1$ exceed u with conditional probability tending to one as $u \to \infty$.

In part (i), $\lim_{u\to\infty} h_I(t_u)=\tau$ and $\lim_{u\to\infty} w(r_I(t_u))=\infty$. Hence $\boldsymbol{\Delta}_I \xrightarrow[u\to\infty]{\mathbb{P}} \mathbf{0}_I$ conditionally on $H_I(u)$. The identity (4.14) and (4.29) imply

$$\mathrm{Law}\left(\frac{\boldsymbol{S}_{\mathcal{C}_I}}{u},\frac{\boldsymbol{Q}_{\mathcal{C}_I}}{u}\Big|H_I(u)\right)\xrightarrow[u\to\infty]{\mathrm{w}}\mathrm{Law}(\boldsymbol{S}_\tau^{(I)},\boldsymbol{Q}_\tau^{(I)}).$$

The limiting margins are atomless at one. Summing the local contributions proves (3.20). For the ratio, write

$$r_I(t_u)^2=2\kappa t_u^2-2(\boldsymbol{\lambda}_I^\top\boldsymbol{\mu}_I)t_u+\boldsymbol{\mu}_I^\top\Sigma_{II}^{-1}\boldsymbol{\mu}_I.$$

Write $\gamma_I=\boldsymbol{\lambda}_I^\top\boldsymbol{\mu}_I$ and $\zeta_I=\boldsymbol{\mu}_I^\top\Sigma_{II}^{-1}\boldsymbol{\mu}_I$. Fix $I_0\in\mathfrak{H}$. If $\gamma_I<\gamma_{I_0}$, then

$$\lim_{u\to\infty}(r_I(t_u)-r_{I_0}(t_u))=\frac{\gamma_{I_0}-\gamma_I}{v_0}>0.$$

By (4.30) and $\lim_{\xi\to\infty} w(\xi)/\xi=1/\tau^2$, for each fixed $b>0$ we have

$$\lim_{\xi\to\infty}\frac{1}{\xi}\log\frac{\overline{F}(\xi+b)}{\overline{F}(\xi)}=-\frac{b}{\tau^2}.$$

Thus the contribution with the smaller score is negligible even after multiplication by any fixed power of t_u. If the scores are tied, then as $u\to\infty$

$$r_I(t_u)-r_{I_0}(t_u)=\frac{\zeta_I-\zeta_{I_0}}{r_I(t_u)+r_{I_0}(t_u)}=\frac{\zeta_I-\zeta_{I_0}}{2v_0t_u}+o(t_u^{-1}),$$

and hence (3.19) with $\lim_{u\to\infty} w(r_{I_0}(t_u))/t_u=v_0/\tau^2$ implies

$$\lim_{u\to\infty}\frac{\overline{F}(r_I(t_u))}{\overline{F}(r_{I_0}(t_u))}=e^{-\frac{\zeta_I-\zeta_{I_0}}{2\tau^2}}.$$

The remaining factors in (4.28) contribute $t_u^{-|I|}$, up to a common factor and a positive constant. Hence the largest scores and then the smallest cardinalities select exactly $\mathfrak{H}$. Consequently, for $I_0\in\mathfrak{H}$ and $I\in\mathfrak{I}_*$

$$\lim_{u\to\infty}\frac{p_I(u)}{p_{I_0}(u)}=\frac{\widehat{A}_I}{\widehat{A}_{I_0}}\mathbb{I}_{\{I\in\mathfrak{H}\}},\qquad \widehat{A}_I=\frac{\tau^{|I|}\exp(-\boldsymbol{\mu}_I^\top\Sigma_{II}^{-1}\boldsymbol{\mu}_I/(2\tau^2))}{(2\pi)^{|I|/2}\sqrt{\det\Sigma_{II}}\prod_{j\in I}\lambda_{I,j}}.$$

These are precisely the weights in (2.8) for covariance $\tau^2\Sigma$, proving the assertion about C_τ.
In part (ii), regular variation with $a<1$ gives $\lim_{u\to\infty} h_I(t_u)=\infty$. Together with $\lim_{u\to\infty} r_I(t_u)w(r_I(t_u))=\infty$, (4.29) yields

$$\frac{\boldsymbol{\Delta}_I}{h_I(t_u)}=O_{\mathbb{P}}\Big((r_I(t_u)w(r_I(t_u)))^{-1/2}\Big)\xrightarrow[u\to\infty]{\mathbb{P}}\mathbf{0}_I\quad\text{conditionally on } H_I(u).$$

Consequently, (4.14) and the log-sum-exp inequalities imply for $\boldsymbol{T}\in\{\boldsymbol{S},\boldsymbol{Q}\}$

$$\mathrm{Law}\left(\left(\frac{\log(T_\ell/u)}{h_I(t_u)}\right)_{\ell\in\mathcal{C}_I}\Big|H_I(u)\right)\xrightarrow[u\to\infty]{\mathrm{w}}\mathrm{Law}\left(\left(\max_{j\in J_{\ell,I}}Z_j^{(I)}\right)_{\ell\in\mathcal{C}_I}\right).$$

The limiting maxima have no atom at zero and their joint positive orthant has positive probability. Hence sums and maxima have the same boundary factors. Summing the local contributions proves (3.21). □

Justification of claims in Section 3.5. In the Gaussian case use the χ_N radial representation with $w(r)\sim r$ and $a=1$. For H defined in (4.31) as $r\to\infty$ we have $H(r)\sim rw(r)/(a+1)\in\mathrm{RV}_{a+1}$, with $a=1$ also in Proposition 3.6(i). For fixed $\boldsymbol{c}\in[0,\infty)^N\setminus\{\mathbf{0}\}$ put $J=\{j:c_j>0\}$, $\sigma_c=\max_{j\in J}\sqrt{\Sigma_{jj}}$ and let $F_{\boldsymbol{c}}$ have survival function $\overline{F}_{\boldsymbol{c}}(u)=|J|^{-1}\sum_{j\in J}\mathbb{P}\{c_je^{X_j}>u\}$. The one-coordinate tail estimate gives $-\log\overline{F}_{\boldsymbol{c}}(e^t)\sim H(t/\sigma_c)$. For $h(u)=u/(\log u)^2$, so that $h(u)\uparrow\infty$ and $h(u)=o(u)$, the marginal Gumbel relation and regular variation yield as $u\to\infty$

$$\frac{\overline{F}_{\boldsymbol{c}}(u-h(u))}{\overline{F}_{\boldsymbol{c}}(u)}\longrightarrow 1,\qquad \frac{\overline{F}_{\boldsymbol{c}}(h(u))^2}{\overline{F}_{\boldsymbol{c}}(u)}\longrightarrow 0,$$

where for the first limit we use $\lim_{u\to\infty} w(u)/u^2 = 0$ and for the second we use $\lim_{u\to\infty} \log h(u)/\log u = 1$ and $a > -1$. Hence splitting the convolution at $h(u)$ implies $\overline{F^{*2}_{\boldsymbol{c}}}(u) \sim 2\overline{F}_{\boldsymbol{c}}(u)$.
For $G \in \mathfrak{R}_N$, [28, Lemma 4.5] gives the gauge $g_G(\boldsymbol{v}) = \sup_{\boldsymbol{p}\in\mathcal{P}_G} \boldsymbol{p}^{\mathsf{T}}\boldsymbol{v}$, where $\mathcal{P}_G \subset [0,\infty)^N$ is bounded, since $\mathbf{0} \notin \overline{G}$. Setting $c_j = g_G(\boldsymbol{e}_j)$, $1 \le j \le N$ and omitting zero coefficients we obtain

$$\max_{j\in J} c_j e^{X_j} \le g_G(e^{\boldsymbol{X}}) \le \sum_{j\in J} c_j e^{X_j}.$$

The one-group cases of Theorem 2.1 and Proposition 3.6 applied to $(X_j + \log c_j)_{j\in J}$ and negligible pairwise intersections imply

$$\mathbb{P}\left\{\max_{j\in J} c_j e^{X_j} > u\right\} \sim \mathbb{P}\left\{\sum_{j\in J} c_j e^{X_j} > u\right\} \sim |J|\overline{F}_{\boldsymbol{c}}(u), \quad u \to \infty.$$

In the elliptical case the required projection retains the Gumbel scaling w, see [15, Lemma 3.5]. The preceding bounds therefore verify [28, Definition 4.6] for $\exp(\boldsymbol{X})$.
For $G \in \mathfrak{R}_d$ set $c_j = g_G(\boldsymbol{e}_\ell)$ for $j \in A_\ell$, $1 \le \ell \le d$ and $J = \{j : c_j > 0\}$. The same argument yields

$$\max_{j\in J} c_j e^{X_j} \le g_G(\boldsymbol{Q}) \le g_G(\boldsymbol{S}) \le \sum_{j\in J} c_j e^{X_j}.$$

The preceding bounds imply as $u \to \infty$

$$\mathbb{P}\{\boldsymbol{S} \in uG\} = \mathbb{P}\{g_G(\boldsymbol{S}) > u\} \sim |J|\overline{F}_{\boldsymbol{c}}(u), \quad \mathbb{P}\{\boldsymbol{Q} \in uG\} = \mathbb{P}\{g_G(\boldsymbol{Q}) > u\} \sim |J|\overline{F}_{\boldsymbol{c}}(u).$$

Since $\overline{F}_{\boldsymbol{c}}$ is subexponential, both $u \mapsto \mathbb{P}\{\boldsymbol{S} \in uG\}$ and $u \mapsto \mathbb{P}\{\boldsymbol{Q} \in uG\}$ are subexponential tails and hence by the arbitrary choice of G we have $\boldsymbol{S}, \boldsymbol{Q} \in \mathcal{S}_{\mathfrak{R}_d}$.
Finally, the dominated variation condition in [28, Proposition 4.15] concerns the coordinate margins of the nonnegative vector. If $0 < a < 1$, the projected log-coordinate X_j has Gumbel scaling $w_j(t) = w((t-\mu_j)/\sigma_j)/\sigma_j \to \infty$. For every fixed $K > 0$ eventually $\log 2 \ge K/w_j(\log u)$ implying

$$\limsup_{u\to\infty} \frac{\mathbb{P}\left\{e^{X_j} > 2u\right\}}{\mathbb{P}\{e^{X_j} > u\}} \le e^{-K}.$$

Letting $K \to \infty$ shows that this ratio tends to zero, which excludes dominated variation and proves the claim made in Section 3.5. □

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